

The On-the-Run Discount*

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Abstract

The on-the-run premium is one of the most widely studied phenomena in fixed-income markets. Contrary to this phenomenon, we document that the most recently issued (on-the-run) Treasury bills trade at a *discount* relative to older bills of comparable maturities. We link this *on-the-run discount* to auction reopenings and short sellers' preference for seasoned bills. The preference arises because short sellers can cover their positions with newly auctioned bills. In line with this mechanism, short-selling demand increases as a bill ages and explains up to 67% of the variation in the on-the-run discount. Price-sensitive investors respond to the discount, but their investments are insufficient to correct it.

Keywords: Convenience premiums, Government debt, Liquidity, Primary dealers, Treasury auctions, Treasuries

JEL: E43, G12, G18, G29

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Introduction

Traditionally, the most recently issued (on-the-run) US Treasury security of a given maturity trades at a premium relative to other Treasuries of comparable maturity. This *on-the-run premium* is one of the most widely studied phenomena in fixed-income markets (e.g., Krishnamurthy, 2002, Goldreich et al., 2005, Vayanos and Weill, 2008, Pasquariello and Vega, 2009, Fleckenstein and Longstaff, 2024). Trading the price difference between on-the-run and more seasoned Treasuries is also a famous hedge fund strategy (e.g., Economist, October 17, 1998; Lowenstein, 2001) and a “textbook” arbitrage opportunity (e.g., Sundaresan, 2009, Veronesi, 2016, Pedersen, 2019).

In this paper, we show that the most recently issued Treasury bills consistently trade at higher yields (i.e., lower prices) than more seasoned bills. Illustrating this *on-the-run discount*, the blue line in Figure 1 plots the yield spread between the first off-the-run 52-week bill and the on-the-run 52-week bill. The grey line plots the yield spread between the seasoned 52-week bill with approximately 26 weeks left to maturity and the newly issued 26-week bill. Because all Treasury bills are widely considered highly liquid and money-like assets (e.g., Duffee, 1996; Krishnamurthy and Vissing-Jorgensen, 2011; Greenwood et al., 2015; Nagel, 2016), a persistent spread between two nearly identical securities is puzzling. Even more puzzling is the sign of the spread, which runs opposite to the established on-the-run premium for coupon-bearing Treasuries.

After exploring the basic properties of the on-the-run discount, we examine the economic forces behind it. We argue that regular auction reopenings make off-the-run bills more attractive to short sellers because they can cover their short positions with the newly issued bills. This demand makes off-the-run bills trade “special” in the repurchase agreement (repo) market, meaning that borrowing cash against them costs significantly less than borrowing

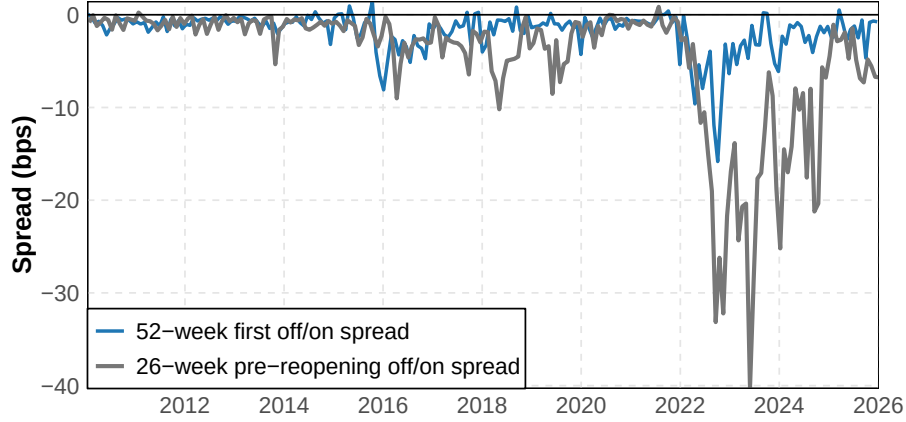


Figure 1: On-the-run discounts for 52-week, and 26-week Treasury bills. The blue line shows the spread between the yield of the first off-the-run bill with an initial maturity of 52 weeks and the current on-the-run bill with 52 weeks to maturity, averaged within each four-week auction cycle. The grey line shows the 26-week pre-reopening off/on spread: the yield of the 52-week bill approaching its 26-week reopening, with 26 to 30 weeks to maturity, minus the yield of the on-the-run 26-week bill, excluding days on which that bill is itself a reopened 52-week security, averaged within each four-week window between consecutive reopening auctions. Both spreads are adjusted for the maturity mismatch between the two securities by subtracting the spread between the OIS rates matching the maturity of each security and are expressed in basis points; negative values mean that the recently supplied bill is cheap. Daily CRSP data from January 2010 to December 2025, 209 observations per series.

against on-the-run bills. The same demand lowers their yields in the cash market. Consistent with this mechanism, we find that short-selling demand for a bill rises in the run-up to its reopening and falls sharply afterward. Moreover, differences in short-selling demand explain up to 67% of the variation in the on-the-run discount.

Comparing off- and on-the-run Treasury bills is particularly straightforward because bills pay no coupons. As a result, their yields are unaffected by accrued interest or coupon differences, making any two bills highly comparable. The principal cash-flow difference between on-the-run bills and their off-the-run counterparts is the remaining time to maturity,

which we adjust for by using liquid overnight index swaps (OIS) with matching maturities. In addition, the regular reopenings allow us to compare on-the-run bills with seasoned bills that mature later. A similar discount in this comparison helps rule out that the on-the-run discount merely reflects a convenience premium in Treasury bills with shorter maturities.

Before exploring the economic forces behind the on-the-run discount, we document four stylized facts about it. First, the spread between off- and on-the-run bills is negatively correlated with the same spread for Treasury notes.¹ Second, its correlation with other common measures of Treasury market liquidity is either insignificant or opposite in sign to that of notes. Third, bid-ask spreads are higher for more seasoned bills, yet they trade at lower yields. Fourth, seasoned bills are substantially more special in the repo market and the difference in repo specialness between seasoned and newly issued bills is strongly correlated with the on-the-run discount.

To examine the drivers of the on-the-run discount, we draw on theories explaining the on-the-run premium. Vayanos and Weill (2008) argue that short sellers concentrate in on-the-run securities because their higher market liquidity makes it easier to close short positions. This demand to borrow on-the-run securities makes them special in the repo market and increases their market value. We apply a similar logic to Treasury bills but argue that short sellers favor off-the-run bills. Once a Treasury bill originally issued with 52 weeks to maturity reaches 26 weeks to maturity, the Treasury reopens it and auctions additional bills with the same CUSIP. Because these bills are identical to the outstanding ones, a short seller can close her position with bills bought at the auction. An upcoming reopening thus plays the same role as market liquidity in Vayanos and Weill (2008). Because off-the-run bills are closer to

¹We calculate the spread between off-the-run and on-the-run Treasuries in the same way for bills and non-bills. As the spread is consistently negative for Treasury bills, we refer to it as “on-the-run discount” to contrast it with the “on-the-run premium” established in previous studies.

their reopening, short sellers' demand to borrow them makes them special and lowers their yields relative to the on-the-run bill.

We construct three proxies of short-selling demand. First, we compute the spread in repo specialness between off- and on-the-run bills. Second, we use the utilization of the Federal Reserve's (Fed's) securities lending facility. The Fed lends Treasury securities from the System Open Market Account (SOMA) to primary dealers, who bid a lending fee for a given security. We measure shorting demand as the ratio of the quantity bid to the lendable supply. This proxy begins in 2019, because SOMA held no Treasury bills before the Fed resumed bill purchases in October 2019. Third, we draw on securities lending data from Markit and define the ratio of securities on loan to active lendable supply as a demand proxy. This proxy is noisier than the Fed measure but covers the full 2010 to 2025 period.

To test our hypothesis, we examine how short-selling demand evolves as a bill ages. Consistent with seasoned bills being more attractive to short sellers, we find that short-selling demand increases with bill age. In addition, we find a sharp discontinuity, with significantly less short-selling demand after the reopening. Short sellers target the very bill that the auction reopens, and their demand disappears once the new supply is delivered. This distinguishes our mechanism from the standard auction cycle, in which dealers short close substitutes to hedge an expected allotment.

Next, we connect short-selling demand to the on-the-run discount. First, linking levels to levels, relative borrowing demand at the Federal Reserve's securities-lending facility explains 30 percent of the variation in the yield spread between 52-week off-the-run and on-the-run bills and 67 percent of the variation in the yield spread between pre-reopening bills and 26-week bills. Second, focusing on changes in the on-the-run discount, we find that a one-percentage-point increase in relative borrowing demand widens both discounts by about 0.08

basis points, with t -statistics between 5.1 and 6.4, and demand changes alone account for 34 and 25 percent of the variation in the two spread changes. The coefficients remain stable after controlling for relative available supply, auction demand, the CDS adjustment, the bill-to-GDP ratio, and other market conditions. Hence, one demand measure prices both ends of the bill lifecycle: when the demand to borrow a bill builds, the newly supplied twin trades cheap.

Broader implication: Term structure of convenience premium.

Having explained the origin of the on-the-run discount, we next examine why the yield differences persist. Many Treasury bill investors value bills for their safety and liquidity and may therefore be relatively insensitive to small yield differences across nearby issues (e.g., Greenwood et al., 2015). Money Market Mutual Funds (MMFs) provide a useful contrast because they compete for small differences in returns (e.g., Kacperczyk and Schnabl, 2013; La Spada, 2018). Although MMFs cannot take short positions, they can reduce their holdings of relatively expensive seasoned bills and increase their holdings of cheaper newly issued bills. Despite regulatory rules that generally favor shorter maturities, we find that MMFs systematically reduce their holdings of 52-week bills as those bills age. This reduction is more pronounced when the on-the-run discount is larger, consistent with MMFs responding to these relative yield differences. However, their portfolio adjustments appear insufficient to eliminate the on-the-run discount.

Primary dealer balance-sheet capacity may also shape the magnitude of the on-the-run discount. Dealers play a central role in intermediating transactions in Treasury cash and repo markets, but this intermediation uses scarce balance-sheet space. The supplementary leverage ratio (SLR) makes Treasury positions costly in terms of dealer balance-sheet capacity (e.g., He et al., 2020), and Duffie (2017) argues that low-risk assets such as Treasury bills

are especially affected by leverage-ratio constraints. Following Du et al. (2018), we examine the on-the-run discount around quarter-end reporting dates, when dealer balance-sheet constraints are particularly pronounced. The discount widens at the end of the calendar quarter, and this widening is significant only during periods when the SLR is in effect. These results are consistent with tighter intermediary constraints amplifying the on-the-run discount.

Our study has two key implications for policymakers. First, because bills trade at lower yields as they approach reopening, our results suggest that the Treasury can benefit from spreading bill auctions across multiple maturities. The introduction of regular 8-week and 17-week bills is consistent with such a finer maturity structure. Our results suggest an additional maturity between 26 and 52 weeks could further lower the Treasury’s borrowing costs. Second, broader central clearing of Treasury transactions (e.g., Duffie, 2020) could reduce the dealer balance-sheet capacity required to intermediate Treasury trades through improved netting. Greater effective intermediation capacity should therefore narrow the on-the-run discount.

1 Related Literature

We contribute to four streams of literature. First, our paper relates to the vast literature examining the pricing of US Treasuries across maturities (Amihud and Mendelson, 1991; Warga, 1992; Gürkaynak et al., 2007; Musto et al., 2018; Fontaine and Garcia, 2012; Hu et al., 2013; Fleckenstein and Longstaff, 2024; among many others) and specifically the pricing of the most recently issued (on-the-run) security compared to the slightly more seasoned (off-the-run) security (e.g., Krishnamurthy, 2002; Fleming, 2003; Goldreich et al., 2005; Vayanos and Weill, 2008; Pasquariello and Vega, 2009). While most of this literature

focuses on coupon-bearing Treasuries,² our focus is on the relative pricing of Treasury bills with small maturity differences. Our contribution to this literature is to show that the premium associated with the most recently issued Treasury flips into a discount because Treasury bills become more attractive to traders as they age.

Second, closely related to the relative pricing of Treasury securities is the concept of repo specialness. Studies focusing on the repo specialness of US Treasuries include Duffie (1996), Jordan and Jordan (1997), D’Amico et al. (2018), and D’Amico and Pancost (2022). These studies have focused on coupon-bearing securities, where the on-the-run Treasury is the special security. We extend this literature to Treasury bills and show that specialness for Treasury bills increases as the security ages. This is the opposite pattern compared to coupon-bearing Treasuries.

Third, our paper contributes to the extensive literature on the underpricing of Treasury auctions (e.g., Cammack, 1991; Spindt and Stolz, 1992; Bikhchandani et al., 2000; Goldreich, 2007; Lou et al., 2013; Beetsma et al., 2016; Herb, 2018; Sigaux, 2024; Fleming et al., 2026). Complementing these auction-price comparisons, Lou et al. (2013) show that the prices of Treasury notes follow a predictable tent-shaped pattern around auctions, dropping in the run-up and recovering afterwards. Whereas Lou et al. (2013) trace the price of a given security over time, we compare the most recently issued Treasury security to older securities at a single point in time, offering a cross-sectional perspective on the cost of auctioning new sovereign debt. While the on-the-run premium was previously viewed as a funding advantage for the Treasury (e.g., Fleming, 2003), the on-the-run discount we document implies an additional cost for the Treasury. The on-the-run discount thus offers a new measure of

²One exception is Fleming (2003), who also examines the on-the-run spread of Treasury bills. However, he finds a noisy pattern that can be attributed to the fact that the on-the-run discount had not materialized in his sample period.

auction underpricing that complements existing time-series measures.

Finally, our paper relates to the literature on the convenience premium of US Treasuries. Traditionally, investors value the safety and liquidity of Treasuries and accept yields below other benchmark rates for the convenience of holding them (Longstaff, 2004; Krishnamurthy and Vissing-Jorgensen, 2012; Greenwood et al., 2015; Nagel, 2016, among many others). More recent evidence suggests that Treasuries can also trade at an “inconvenience premium” with yields above other benchmark rates (Du et al., 2018; Duffie, 2020; He et al., 2020; Klingler and Sundaresan, 2023; Jiang et al., 2025). While the on-the-run discount is not itself a measure of (in)convenience premiums, it points to similar tensions in the Treasury market arising from the recent increase in Treasury supply and elevated primary dealer constraints.

2 Background and Data

In this section, we present the institutional background and describe the data underlying our analysis. We begin with an overview of the Treasury bill market and explain how we construct the on-the-run discount. Afterwards, we introduce three proxies for short selling demand.

2.1 The Treasury Bill Market

Treasury bills are zero-coupon securities with original maturities of up to 52 weeks. They are typically considered safe and liquid assets with money-like features (e.g., Greenwood et al., 2015, Nagel, 2016). One difference between Treasury bills and coupon-bearing Treasuries (which have initial maturities ranging from 2 to 30 years) is that the market liquidity of

coupon-bearing securities drops sharply as the security ages (e.g., Fleming, 2003; Barclay et al., 2006; Pasquariello and Vega, 2009; Goyenko et al., 2011; Musto et al., 2018, Brain et al., 2018). By contrast, the trading volume of Treasury bills does not drop as sharply as the security ages (Brain et al., 2018). Two potential reasons for this continued market liquidity of seasoned Treasury bills are (i) the Treasury auctioning mechanism, which often involves reopening existing Treasury bills and (ii) their relatively short time to maturity and increased money-likeness as bills age.³

To elaborate on the first point, consider the 52-week Treasury bill, which has been issued every 4 weeks since its reintroduction in June 2008. When the original 52-week bill reaches a remaining maturity of 26 weeks, the Treasury reopens the same security through a 26-week bill auction. Trading activity for a given security typically peaks around its auction (e.g., Lou et al., 2013; Fleming et al., 2024), which can boost the market liquidity of the Treasury bill (Pasquariello and Vega, 2009). Therefore, the reopening of seasoned bills helps maintain their high market liquidity. Coupon-bearing Treasuries are also reopened, often shortly after original issuance. However, these reopenings typically add supply to the current on-the-run issue, whereas bill reopenings can reintroduce an aging security as the on-the-run issue in a shorter bill tenor.⁴

In our analysis, we focus on this reopening cycle for 52-week Treasury bills as their 4-week auction cycle closely resembles the issuance pattern of coupon-bearing securities. We first compare the newly issued 52-week bill with the first off-the-run bill, issued four weeks earlier. In addition, we compare the 52-week bill in the weeks before its reopening with the

³The shorter the time to maturity of a Treasury bill, the lower its sensitivity to changes in interest rates and hence its price volatility. This lower volatility translates into lower bid-ask spreads (e.g., Ho and Stoll, 1981; Amihud and Mendelson, 1991).

⁴Even though coupon-bearing Treasuries can also be reintroduced as a shorter-tenor issue, the time between issuance and reopening is substantially longer (i.e., years, not months).

on-the-run 26-week bill.

We obtain prices for Treasury bills and notes from CRSP and supplement these data with bid-ask spreads and high-frequency changes around auctions obtained from GovPX.⁵ To identify on-the-run and off-the-run securities, we obtain detailed information on Treasury auctions from TreasuryDirect. Panel A of Table 1 provides summary statistics of the Treasury bill yields. The average 52-week on-the-run yield is 1.52% and two basis points higher than the average yield of the first 52-week off-the-run bill. Fleming (2003) observed a similar pattern and attributed it to the upward-sloping term structure of money market yields. This explanation does not align with the lower yield of the pre-reopening off-the-run bill (1.44%) compared to the yield of the on-the-run 26-week bill (1.48%).

2.2 Quantifying the On-the-Run Discount

Comparing on- and off-the-run Treasury bills is straightforward because bills are zero-coupon securities. In contrast to coupon-bearing Treasuries, the yields of Treasury bills are unaffected by differences in coupon rates, accrued interest, or coupon reinvestment. The main difference between different bills is their remaining time to maturity. This maturity gap is about four weeks for 52-week bills and ranges from one to four weeks for the 26-week bill. Hence, isolating the on-the-run component requires adjusting for the slope of the risk-free term structure. We do so by subtracting the difference in maturity-matched benchmark rates from the yield difference between the off-the-run and on-the-run bills.

Let $y_{i,t}$ denote the yield of security i at time t . We set $i = 0$ for the on-the-run security and $i = k$ for the k -th off-the-run security, where $k < 0$ indicates that the security will be reopened (and turn on-the-run again) after $|k|$ auction cycles. Let $z_{i,t}$ denote the benchmark

⁵We use two data sources for bid-ask spreads because Adrian et al. (2017) show that bid-ask spreads are often stale for on-the-run securities in the CRSP database.

Table 1: Summary statistics. Panel A reports average unadjusted CRSP midpoint yields for the four bills that form the two on-the-run pairs: the on-the-run 52-week bill and the first off-the-run bill, and the pre-reopening off-the-run bill, a 52-week bill with 26 to 30 weeks to maturity, and the on-the-run 26-week bill. Panel B reports the maturity-gap adjustments in basis points: $z_{1,t} - z_{0,t}$ for the first off/on pair and $z_{pre,t} - z_{26w,t}$ for the reopening pair, where $z_{i,t}$ is the benchmark yield at bill i 's remaining maturity. The OIS rows are the adjustments subtracted in the two spreads; the GSW and OIS-plus-CDS rows are alternatives, and the last column reports their correlation with the OIS adjustment. Panel C reports average repo specialness, the benchmark overnight repo rate minus the security-specific repo rate, available from August 2012. The on-the-run and first off-the-run rows average daily observations within each four-week auction cycle; the pre-reopening and 26-week rows average within each four-week reopening window. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

<i>Panel A. Auction-cycle average unadjusted CRSP midpoint yields (%), 2010–2025</i>							
	Mean	SD	P25	Median	P75	<i>N</i>	
On-the-run yield	1.52	1.75	0.16	0.53	2.55	209	
First off-the-run yield	1.50	1.75	0.14	0.47	2.54	209	
Pre-reopening off-the-run yield	1.44	1.78	0.09	0.38	2.42	209	
On-the-run 26-week yield	1.48	1.84	0.10	0.41	2.45	209	
<i>Panel B. Auction-cycle average maturity-gap adjustments (bps), 2010–2025</i>							
	Mean	SD	P25	Median	P75	<i>N</i>	Corr. with OIS
$z_{1,t} - z_{0,t}$ (OIS)	−0.29	2.28	−1.77	−0.39	0.77	209	
$z_{1,t} - z_{0,t}$ (GSW)	−0.25	2.23	−1.61	−0.42	0.43	209	0.96***
$z_{1,t} - z_{0,t}$ (OIS + CDS)	−0.50	2.33	−1.96	−0.59	0.33	209	0.94***
$z_{pre,t} - z_{26w,t}$ (OIS)	0.24	1.50	−0.07	0.17	0.80	209	
$z_{pre,t} - z_{26w,t}$ (GSW)	−0.06	1.49	−0.58	−0.17	0.59	209	0.87***
$z_{pre,t} - z_{26w,t}$ (OIS + CDS)	0.37	1.49	−0.07	0.28	0.91	209	0.96***
<i>Panel C. Auction-cycle average repo specialness (bps), 2012–2025</i>							
	Mean	SD	P25	Median	P75	<i>N</i>	
On-the-run specialness	−0.53	6.72	−4.38	−1.91	0.95	169	
First off-the-run specialness	8.96	20.60	−1.89	2.61	10.97	168	
Pre-reopening off-the-run specialness	49.54	92.04	5.39	15.21	41.56	173	
On-the-run 26-week specialness	−2.13	3.86	−4.63	−2.48	0.20	170	

rate with the same remaining maturity as bill i . To calculate the on-the-run spread, we use the benchmark-adjusted yield difference between the k -th off-the-run bill and the on-the-run bill:

$$Spread_{0,t}^k = y_{k,t} - y_{0,t} - (z_{k,t} - z_{0,t}), \quad (1)$$

which is lower when the on-the-run security is cheaper than the off-the-run security. For the first spread in our analysis pair, we focus on 52-week on-the-run bills and $k = 1$. For the second spread, we focus on 26-week on-the-run bills and $k = -1$.

We refer to these spreads as on-the-run *discounts* when they are negative. For coupon-bearing securities, this spread is typically positive and therefore often referred to as an on-the-run premium. In the empirical analysis, we average daily spreads within each auction cycle (i.e., between the auction date of a new 52-week bill and the day before the next 52-week bill auction) and, for the second pair, within each reopening window between consecutive 26-week reopening auctions, excluding the days on which the on-the-run 26-week bill is itself a reopened 52-week bill. This helps mitigate short-lived fluctuations around auction days (e.g., Lou et al., 2013 or Fleming et al., 2026) and allows us to focus on the driver of the overall spread.

Our preferred proxy for z comes from overnight-index swaps (OIS). In an OIS with 18 or fewer months to maturity, two parties exchange a fixed overnight rate against the average overnight rate during the lifetime of the swap.⁶ OIS contracts are liquid and trade at granular

⁶The floating leg is based on the compounded overnight rate and the daily overnight rates therefore enter through a geometric rather than arithmetic average. Traditionally, the predominant floating rate in OIS is the effective federal funds rate (FFR). This changed after the transition from London Interbank Offered Rate (LIBOR) to Secured Overnight Financing Rate (SOFR) in the U.S. (e.g., Klingler and Syrstad, 2021). However, to obtain a consistent time series, we use FFR OIS for our main adjustment and show in the appendix that our results remain virtually unchanged when using SOFR OIS for the later period instead.

monthly maturities between six and twelve months. Because we use z only to adjust for the slope of the yield curve, the precise choice of z is less important than in other studies that, for example, examine the (in)convenience premium of Treasuries. As a robustness check, we also construct spreads using the smooth yield curve estimated by Gürkaynak et al. (2007).⁷

An additional concern with comparing Treasury bills with different maturities is differences in credit risk. As noted by Dittmar et al. (2026), political conflict over the debt ceiling has led to increased credit risk of Treasuries. Such credit risk could affect the on-the-run discount if the off-the-run bill matured before a debt ceiling deadline while the on-the-run bill matured after the deadline. To capture such differences in credit risk, we follow Fleckenstein and Longstaff (2024) and add the interpolated sovereign CDS premium to z .⁸

Panel B of Table 1 shows that the maturity adjustment spreads, $z_{k,t} - z_{0,t}$, are small on average and highly correlated across different benchmark specifications. All relevant correlations are at least 0.87, suggesting that the choice of spread adjustment has, at best, a second-order effect on the on-the-run discount. We therefore focus on the OIS adjustment and later show that the spread in CDS premiums is not significantly related to the on-the-run discount.⁹

⁷This curve has been used by, among others, Goldreich et al. (2005) and Nagel (2016) to adjust for differences in coupon and maturity structure between on-the-run and off-the-run notes. However, the Gürkaynak et al. (2007) curve is based on coupon-bearing securities with more than one year to maturity and therefore less suitable to adjust bill spreads.

⁸While multiple studies emphasize the credit risk of the US Treasury (e.g., Augustin et al., 2021, Dittmar et al., 2026), Klingler and Lando (2018) argue that sovereign CDS are not a clean proxy of credit risk. This is less of a concern for our analysis because we only need the spread between CDS of different maturities for our adjustment.

⁹Because debt-limit risk is concentrated around particular payment dates, Table IA.8 in the Internet Appendix also excludes acute debt-limit periods and bill pairs whose maturity dates straddle projected X-dates. Because a 52-week on-the-run pair is priced about a year before either bill matures, no pair is priced during a period of acute debt-limit concern while either bill matures within six weeks of the projected X-date, so these exclusions are conservative. The average $Spread_0^1$ is -1.77 basis points in the baseline sample, -1.81 basis points after excluding entire auction cycles that overlap acute periods, and -1.82 basis points after additionally excluding maturity-straddling pairs. The largest discounts occur in 2022, outside debt-limit episodes.

Figure 1 shows the time series of the two spreads. Both are negative in almost every period. The 52-week first off/on spread averages -1.8 basis points and is negative in 92% of auction cycles; the 26-week pre-reopening off/on spread averages -4.6 basis points and is negative in 99% of reopening windows. The discounts are largest in 2022 and 2023, when the first off/on spread reaches -16 basis points and the pre-reopening spread -40 basis points. Both series are persistent, with first-order autocorrelations of 0.63 and 0.86. Hence, in our empirical analysis, we focus on first differences in the spreads when studying their relationship to other variables.

2.3 Repo Specialness and Securities Lending

To compare the repo specialness of on- and off-the-run bills, we rely on granular security-level special repo data from the Depository Trust & Clearing Corporation (DTCC). The data cover bilateral delivery-versus-payment repos cleared by DTCC’s Fixed Income Clearing Corporation, the interdealer market in which a dealer that needs a particular Treasury security borrows it from another dealer. For each security and day, DTCC reports the number of trades, the volume, and the volume-weighted repo rate, from 2011 on. The average total daily overnight repo volume captured in the DTCC data is \$500 billion during our sample period, representing a substantial fraction of the overall US repo market. To construct a measure of repo specialness, we compute the difference between the market-wide Treasury repo rate for general collateral repos and the repo rate for security i on the same date. Our preferred proxy for market-wide repo rates is the secured overnight financing rate (SOFR), which captures the cost of overnight borrowing collateralized with US Treasuries (see Klingler and Syrstad (2021) for more details). Because SOFR is only constructed from 2014 on, we use the weighted average of the triparty repo rate and the general collateral

financing (GCF) rate before then.¹⁰

Panel C of Table 1 contains summary statistics for repo specialness for the four bills. The two on-the-run bills exhibit virtually no specialness: the spread between the benchmark repo rate and the special repo rate averages -0.5 basis points for the 52-week bill and -2.1 basis points for the 26-week bill. In contrast, the spread is 9 basis points for the first off-the-run bill and 49.5 basis points for the pre-reopening off-the-run bill. In both pairs the seasoned bill is the special one, and by far the most so in the weeks before its reopening. This pattern is consistent with Panel A, which shows that the seasoned bills have lower yields than their on-the-run counterparts.

A special repo rate shows that a bill is in demand from borrowers. To observe borrowing directly, we use two sources of securities lending data. The first is the Federal Reserve’s securities lending facility, through which the Federal Reserve lends Treasury securities held in the System Open Market Account (SOMA) to primary dealers in a daily auction. For each security and operation date, the data report the par amount submitted by dealers and the par amount available for lending, and we define $BD_{i,t}^{Fed}$ as par submitted divided by par available, in percent.¹¹ Because dealers bid at the facility only when a security cannot be obtained more cheaply in the market, the measure captures excess demand for a specific security. Usable coverage of 52-week bills begins in late 2019, when the Federal Reserve expanded its holdings of these securities through reserve-management purchases.

The second source is S&P Global Securities Finance (formerly IHS Markit), which col-

¹⁰This variable captures most transactions that are also included in the calculation of SOFR. Over the overlapping period, this benchmark is nearly perfectly correlated with SOFR, with a correlation of 0.9999.

¹¹The daily lifecycle and reopening-discontinuity exercises use reported security-operation observations, which are days with bids. For the auction-cycle measures, we also include eligible operation days without bids and assign them zero demand before averaging. We treat a security as eligible from its first appearance in the lending results or, for bills issued after October 2019, from its issue date, as nearly all such bills appear in the results within days of issuance.

lects daily records from the custodians and agent lenders that lend securities on behalf of institutional investors. For each security and day, we compute the short-loan share, $BD_{i,t}^{Mkit}$, as the quantity on loan to short sellers, net of loans made for financing purposes, divided by the quantity that lenders actively make available, in percent; days with lendable inventory but no loans count as zero. The two measures complement each other. The Federal Reserve measure records bids at a backstop and is the sharper signal of scarcity, but it exists only for securities the Federal Reserve holds and only from late 2019. The Markit measure records the stock of loans outstanding in lending programs, which do not include dealer-to-dealer borrowing through repo, but it is available for every bill from 2010. Across security-days, the two are positively correlated, with a correlation of 0.6.

3 Motivating Evidence

In this section, we document several stylized facts about the on-the-run discount and its relation to Treasury market liquidity and repo specialness.

3.1 Correlation with Other Measures

We use the term “on-the-run discount” for spreads between off- and on-the-run bills because the spread is predominantly negative. This terminology helps distinguish our results from the spread between off- and on-the-run securities that has been previously studied for coupon-bearing securities and that is predominantly positive. To highlight the qualitative differences between bills and coupon-bearing securities, Table 2 shows the correlation between spreads of off- and on-the-run securities of different tenors. In the table, we focus on monthly averages of the spreads and analyze first differences to avoid spurious correlations due to non-stationary

time series.¹²

The correlations indicate a distinct bill-market phenomenon. Changes in the 26-week and 52-week bill spreads are positively correlated, suggesting a common component in relative bill pricing. By contrast, bill spreads are negatively correlated with 2-year note on-the-run spreads and only weakly related to 10-year note spreads, while the note spreads are positively correlated with each other. The bill spreads also do not line up with standard Treasury liquidity or stress measures in the same way as note spreads: their correlations with the note-bill spread (Amihud and Mendelson, 1991), the noise measure (Hu et al., 2013), and the MOVE index are either weak or of the opposite sign. Taken together, these patterns suggest that the on-the-run discount is distinct from the previously documented on-the-run premium.

Table 2: Correlation between on-the-run discounts and other measures. This table reports pairwise correlations of monthly changes among the two bill on-the-run discounts, the on-the-run spreads of 2- and 10-year notes, and other Treasury-market measures. $Spread_{26w}$ is the 26-week pre-reopening off/on spread; $Spread_{52w}$, $Spread_{2y}$, and $Spread_{10y}$ are the OIS-adjusted first off/on spreads for 52-week bills and 2- and 10-year notes; Note/bill is the note-bill spread of Amihud and Mendelson (1991); Noise is the noise measure of Hu et al. (2013); MOVE is the MOVE index. All series are monthly averages of daily data, first-differenced, January 2010 to December 2025. p -values in brackets; ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	$\Delta Spread_{52w}$	$\Delta Spread_{2y}$	$\Delta Spread_{10y}$	Δ Note/bill	Δ Noise	Δ MOVE
$\Delta Spread_{26w}$	0.15** [0.04]	0.03 [0.66]	-0.09 [0.21]	-0.23*** [0.00]	-0.12 [0.10]	0.03 [0.65]
$\Delta Spread_{52w}$		-0.19*** [0.01]	0.01 [0.84]	-0.13* [0.08]	-0.07 [0.32]	-0.10 [0.17]
$\Delta Spread_{2y}$			0.16** [0.03]	0.28*** [0.00]	0.32*** [0.00]	0.23*** [0.00]
$\Delta Spread_{10y}$				0.15** [0.04]	0.15** [0.03]	0.22*** [0.00]

¹²Table IA.1 in the Internet Appendix shows that the results are qualitatively similar when focusing on levels instead of first differences.

3.2 Market Liquidity of Bills

One natural explanation for any on-the-run spread is differences in market liquidity. For coupon-bearing securities, the most recently issued security is typically more liquid than older issues and therefore trades at a lower yield (e.g., Barclay et al., 2006; Pasquariello and Vega, 2009). If the on-the-run discount for bills reflected the same market liquidity channel in reverse, the on-the-run bill would have to be less liquid than seasoned bills. To evaluate this possibility, we construct bid-ask spread differentials analogously to the on-the-run yield spread by subtracting the bid-ask spread of the on-the-run bill from the bid-ask spread of the off-the-run bill. As noted before, given the evidence in Adrian et al. (2017), we supplement the bid-ask spreads from CRSP with bid-ask spreads from GovPX.

Table 3 shows that the bid-ask spread differentials are positive in every case, that is, in the direction opposite to what a liquidity explanation of the on-the-run discount requires. For the 52-week first off/on pair, the average differential is 0.09 basis points of price in CRSP data and 0.45 in GovPX data; for the 26-week pre-reopening pair it is 0.52 and 1.07, so the bill approaching its reopening is clearly the less liquid of the two. Moreover, the correlations between changes in the bid-ask differentials and changes in the spreads are negative, weak in the CRSP data and significant in the GovPX data. That is, the seasoned bill becomes relatively less liquid precisely when its price advantage over the on-the-run bill grows. Thus, market liquidity differences do not explain the on-the-run discount and, if anything, work in the opposite direction.

3.3 Special Repo Rates

We next investigate if differences in repo specialness are related to the on-the-run discount. Repo markets provide a direct signal of how difficult individual Treasury securities are to

source (e.g., Duffie, 1996, Jordan and Jordan, 1997, D’Amico et al., 2018, D’Amico and Pan-cost, 2022). We construct repo-specialness differentials as the on-the-run repo rate minus the off-the-run repo rate. Because a lower repo rate indicates greater specialness, positive values indicate that the off-the-run bill is more special than the on-the-run bill. If funding-market scarcity is relevant for the on-the-run discount, then these repo-specialness differentials should be related to the yield spread between older and newly issued bills.

Table 3 is consistent with this prediction. The specialness differential is positive on average in both pairs, 8.5 basis points for the first off/on pair and 55 basis points for the pre-reopening pair, so the seasoned bill is the more special one, and by a wide margin when it approaches its reopening. Changes in the differential are also strongly negatively correlated with changes in the spread, with correlations of -0.46 and -0.65 . When the seasoned bill becomes more special, its yield falls further below the on-the-run bill’s. These correlations show that the cash-market discount moves with the relative cost of borrowing seasoned bills in the repo market.

4 The Mechanism behind the On-the-run Discount

The evidence so far suggests that differences in transaction costs cannot explain the on-the-run discount. By contrast, there is a strong link between the on-the-run discount and the specialness of the off-the-run security in the repo market. In this section, we revisit theories explaining the on-the-run premium and discuss how the market structure of the Treasury bill market can give rise to an on-the-run discount. Afterwards, we test whether the Treasury bill reopening cycle gives rise to the borrowing demand that underlies the on-the-run discount.

Table 3: Market liquidity, repo specialness, and the on-the-run discounts. This table reports means, medians, and correlations for benchmark-adjusted bill spreads, repo-specialness spreads, and bid-ask spread differentials, averaged within four-week auction cycles (Panel A) or reopening windows (Panel B). Panel A compares the on-the-run 52-week bill with the first off-the-run bill. Panel B reports the 26-week pre-reopening off/on spread: the 52-week bill approaching its 26-week reopening against the on-the-run 26-week bill. Repo-specialness spreads are the on-leg repo rate minus the off-leg repo rate, in basis points, so positive values indicate that the seasoned bill is more special. Bid-ask differentials are off-leg minus on-leg spreads in basis points of price, from CRSP and from GovPX. The correlation column uses cycle-to-cycle changes of each variable against the change in the spread. p -values from pairwise correlation tests; ***, **, and * indicate significance at the 1%, 5%, and 10% levels. The sample period is 2010–2025.

<i>Panel A. The 52-week first off/on spread, 2010–2025</i>					
	Mean	Median	N	Corr. with $\Delta Spread$	p -value
$Spread_{52w}$	−1.77	−1.05	209		
$Repo\ Specialness\ Spread_{52w}$	8.53	2.77	189	−0.46***	0.00
$Bid\text{-}ask\ Spread_{52w}$	0.09	0.11	209	−0.10	0.17
$Bid\text{-}ask\ Spread_{52w}\ (GovPX)$	0.45	0.31	208	−0.28***	0.00

<i>Panel B. The 26-week pre-reopening off/on spread, 2010–2025</i>					
	Mean	Median	N	Corr. with $\Delta Spread$	p -value
$Spread_{26w}$	−4.63	−1.86	209		
$Repo\ Specialness\ Spread_{26w}$	55.01	17.00	189	−0.65***	0.00
$Bid\text{-}ask\ Spread_{26w}$	0.52	0.39	209	−0.10	0.16
$Bid\text{-}ask\ Spread_{26w}\ (GovPX)$	1.07	0.67	208	−0.46***	0.00

4.1 Hypothesis

Explanations for the on-the-run premium hinge on the superior market liquidity of the on-the-run security (Krishnamurthy, 2002, Vayanos and Weill, 2008, Pasquariello and Vega, 2009). In Vayanos and Weill (2008), short sellers prefer the liquid on-the-run security because its liquidity allows them to close their positions easily. The demand by short sellers further increases the market liquidity and repo specialness of the on-the-run security, in turn increasing its price.

Bill reopenings reverse which security attracts short sellers. Treasury bills continue to trade actively as they age (e.g., Fleming, 2002, Barclay et al., 2006, Brain et al., 2018), and an original 52-week bill is eventually reopened as a 26-week bill. As the reopening approaches, traders may establish short positions in the bill for a variety of reasons, including to manage expected auction exposure (e.g., Lou et al., 2013, Beetsma et al., 2016, Klingler and Sundaresan, 2023, Fleming et al., 2026). The reopening makes such positions particularly attractive because it supplies additional amounts of the same CUSIP on a known settlement date, allowing traders to cover their short positions with the newly delivered securities. Borrowing demand for the seasoned bill can therefore rise before reopening even though its cash-market liquidity does not improve.

This mechanism yields three empirical predictions.

- (i) Borrowing demand for a given Treasury bill increases as the security approaches its reopening date and falls once the additional securities have become deliverable.
- (ii) The on-the-run discount widens when borrowing demand for the off-the-run bill increases relative to borrowing demand for the on-the-run bill.
- (iii) Among off-the-run bills, the bills closer to their reopening date should trade at a lower yield and therefore a higher convenience premium.

4.2 Short-Selling Demand

To test these predictions, we use three proxies for security-specific short-selling demand. Repo specialness provides an indirect measure of demand to borrow a particular security. Our two more direct measures come from securities-lending data. The primary measure, $BD_{i,t}^{Fed}$, is the ratio of dealers' submitted bids for security i in the Federal Reserve's securities-lending operations to the amount available for lending. A complementary measure, $BD_{i,t}^{Mkit}$,

is the ratio of the quantity of security i on loan to its actively lendable quantity in the Markit data.¹³ We use OIS-adjusted Treasury richness to examine how this borrowing pressure is reflected in cash-market prices.

We first examine how borrowing pressure evolves as an original 52-week bill approaches its 26-week reopening. Figure 2 presents the three proxies in age bins from the initial auction through the first four weeks after the reopening settlement. Panel (a) shows repo specialness, Panel (b) borrowing demand in the Federal Reserve’s securities-lending operations, and Panel (c) the Markit short-loan share. All three increase as the reopening approaches, reach their highest values in the final pre-settlement bin, and fall sharply once the reopening supply is delivered and the bill becomes the new on-the-run 26-week bill. The similar pattern across all three proxies indicates a common lifecycle in security-specific borrowing pressure.

Figures 2 and 4 provide descriptive evidence that borrowing pressure and cash-market richness follow the reopening cycle. We next sharpen the first prediction by exploiting the timing of reopening settlement. Because the newly auctioned securities cannot be used to cover short positions until they are delivered, borrowing pressure should fall discontinuously when the additional supply becomes available. We test this prediction using daily observations around each bill’s reopening settlement.

For each original 52-week bill, we use observations from its initial issue date through four weeks after its 26-week reopening auction and estimate

$$Y_{i,t} = \beta_1 Distance_{i,t} + \beta_2 Distance_{i,t} \times \mathbb{1}(\text{Reopened})_{i,t} + \beta_3 \mathbb{1}(\text{Reopened})_{i,t} + \lambda_t + \epsilon_{i,t}, \quad (2)$$

¹³The Markit measure covers securities lending through participating custodians and agent lenders but does not capture important channels of Treasury borrowing, particularly dealer-to-dealer borrowing through repo. In addition, variation in contributor coverage and actively lendable supply introduces measurement error into the short-loan share. Consistent with the narrower coverage, Table IA.2 in the Internet Appendix shows that the quantity of Treasury bills observed on loan in Markit is relatively small.

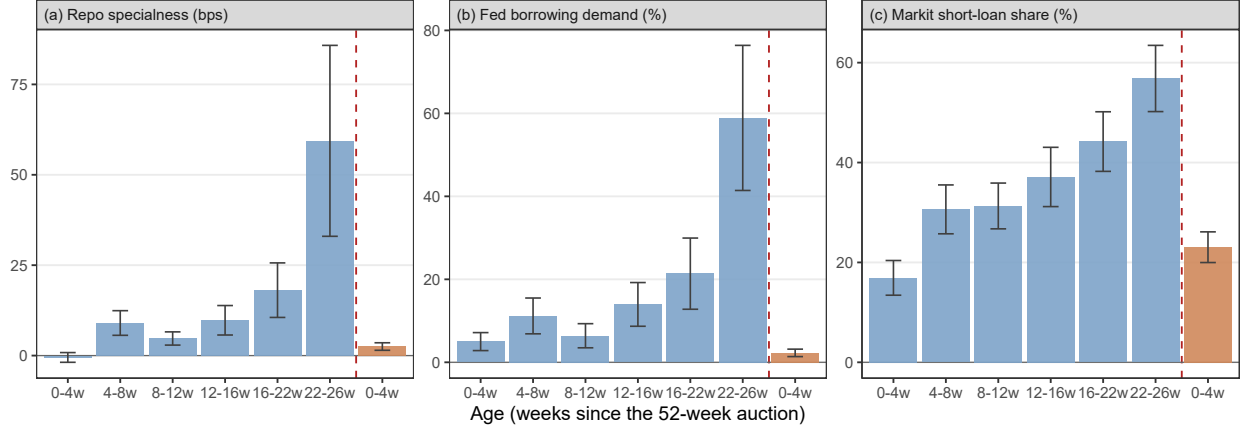


Figure 2: Borrowing demand and repo specialness before and after the 26-week reopening of a 52-week Treasury bill. Each bar first averages daily observations within a bill and age bin and then averages across original 52-week bills; error bars show 95% confidence intervals based on Newey–West standard errors. Panel (a) plots repo specialness, measured as the benchmark overnight repo rate minus the security-specific repo rate, in basis points. Panel (b) plots $BD_{i,t}^{Fed}$, par submitted relative to par available in the Federal Reserve’s securities-lending operations, in percent. Panel (c) plots $BD_{i,t}^{Markit}$, short-loan quantity relative to actively lendable quantity in the Markit data, in percent. Age is measured in weeks from the 52-week auction; every day before the settlement of the bill’s 26-week reopening belongs to the 22–26 week bin. The orange bar covers the first four weeks from the settlement, when the bill is the new on-the-run 26-week bill, and the red dashed line marks the settlement. The samples are bills whose window from the auction to four weeks after the settlement lies within the data: 2012–2025 for specialness (166 bills), December 2019 to 2025 for the Federal Reserve measure (72 bills), and 2010–2025 for the Markit measure (202 bills).

where $Y_{i,t}$ is repo specialness, $BD_{i,t}^{Fed}$, or $BD_{i,t}^{Mkit}$. $Distance_{i,t}$ is the number of days relative to the settlement of bill i 's reopening and is negative before settlement. The indicator $\mathbb{1}(\text{Reopened})_{i,t}$ equals one beginning on the settlement date, and λ_t denotes date fixed effects. The two distance terms allow separate linear trends before and after settlement, while β_3 measures the discontinuity when the reopening supply becomes deliverable.

Table 4: Discontinuity at the delivery of the reopening supply. This table reports daily panel regressions for original 52-week Treasury bills from each bill's issue date to four weeks after its 26-week reopening auction, estimated from Equation (2). Column (1) uses repo specialness as the dependent variable, measured as the benchmark overnight repo rate minus the security-specific repo rate, in basis points, 2012–2025. Column (2) uses $BD_{i,t}^{Fed}$, borrowing demand for the security in the Federal Reserve's securities-lending operations, measured as the ratio of par submitted to par available on operation days with bids, in percent, October 2019 to December 2025. Column (3) uses $BD_{i,t}^{Mkit}$, the Markit short-loan share, measured as short-loan quantity relative to actively lendable quantity on days with lendable inventory, in percent, 2010–2025. $Distance_{i,t}$ is the number of days between date t and the settlement of bill i 's reopening, negative before it, and $\mathbb{1}(\text{Reopened})_{i,t}$ equals one from the settlement day, so its coefficient is the jump at the delivery of the reopening supply. All specifications include date fixed effects. The numbers in parentheses are t -statistics based on standard errors clustered by date and CUSIP. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Specialness (1)	$BD_{i,t}^{Fed}$ (2)	$BD_{i,t}^{Mkit}$ (3)
$\mathbb{1}(\text{Reopened})_{i,t}$	−42.56*** (−8.67)	−44.08*** (−9.09)	−38.75*** (−15.89)
$Distance_{i,t} \times \mathbb{1}(\text{Reopened})_{i,t}$	−0.65*** (−4.97)	−0.67*** (−5.68)	−0.01 (−0.16)
$Distance_{i,t}$ (days)	0.35*** (8.33)	0.33*** (9.13)	0.24*** (12.38)
Date FE	Yes	Yes	Yes
Sample	2012–2025	2019–2025	2010–2025
Observations	20,635	7,717	30,010
R ²	0.28	0.36	0.33
Adjusted R ²	0.15	0.21	0.22

Table 4 shows a large and statistically significant decline in all three outcomes at the reopening settlement. Relative to the fitted pre-settlement trend, repo specialness falls by 43 basis points, Federal Reserve borrowing demand by 44 percentage points, and the Markit short-loan share by 39 percentage points. Before settlement, specialness rises by 0.35 basis points per day, Federal Reserve borrowing demand by 0.33 percentage points per day, and the Markit short-loan share by 0.24 percentage points per day as the reopening approaches. For specialness and Federal Reserve borrowing demand, the interaction coefficients show that this buildup reverses after settlement.

Because borrowing pressure accelerates during the final weeks before reopening, the linear specification may understate the discontinuity. Allowing for quadratic trends in Table IA.6 increases the estimated settlement declines to 75 basis points for specialness and 69 percentage points for Federal Reserve borrowing demand. Table IA.7 addresses the possibility that the ratios fall mechanically because lendable quantities increase at settlement. Holding each denominator at its final pre-settlement value reduces the Federal Reserve decline only modestly, from 44 to 38 percentage points. The Markit decline falls from 39 to 17 percentage points, indicating that its baseline decline partly reflects the increase in lendable inventory but that short-loan quantity also falls at settlement. Together with Figure 4, these estimates link the lifecycle of borrowing pressure, repo specialness, and cash-market richness to the reopening cycle rather than to a smooth function of remaining maturity.

4.3 The On-the-Run Discount and Short-Selling Demand

The second prediction links relative borrowing demand to the relative cash prices of seasoned and newly issued bills. For each pair p , we define the Federal Reserve and Markit borrowing-demand spreads as borrowing demand for the seasoned security minus borrowing demand

for the newly issued one, averaging both measures over the same auction cycle or reopening window used to construct the corresponding yield spread. We define the repo-specialness spread analogously. Thus, a positive borrowing-demand or specialness spread means that the seasoned bill is more heavily demanded by borrowers or is more special. Because $Spread_{p,t}$ is the OIS-adjusted yield of the seasoned bill minus that of the newly issued bill, the mechanism predicts a negative relationship between the yield spread and each measure of relative borrowing pressure.

Figure 3 illustrates this relationship in levels using $BD^{Fed} Spread_{p,t}$, the Federal Reserve borrowing-demand spread. Relative borrowing demand explains 30% of the variation in the 52-week first off/on spread in Panel (a) and 67% of the variation in the 26-week pre-reopening spread in Panel (b). In both cases, greater borrowing demand for the seasoned bill is associated with a more negative on/off spread, meaning that the seasoned bill trades richer relative to the newly issued bill and the on-the-run discount is larger.

Because both the yield spreads and the measures of relative borrowing pressure are persistent, our formal tests use window-to-window changes. We estimate the specification separately using the Federal Reserve borrowing-demand spread, the Markit borrowing-demand spread, and the repo-specialness spread. In the controlled specifications, we retain an auction-based measure of relative available supply. For each bill, AS_i is the par amount awarded to primary dealers at the auction that initially issued the bill, which proxies for the quantity initially placed with intermediaries. We define $AS Spread_{p,t}$ as the corresponding difference in log available supply. Because AS_i is fixed at issuance, this control captures persistent differences in initial intermediated float rather than contemporaneous lendable supply.

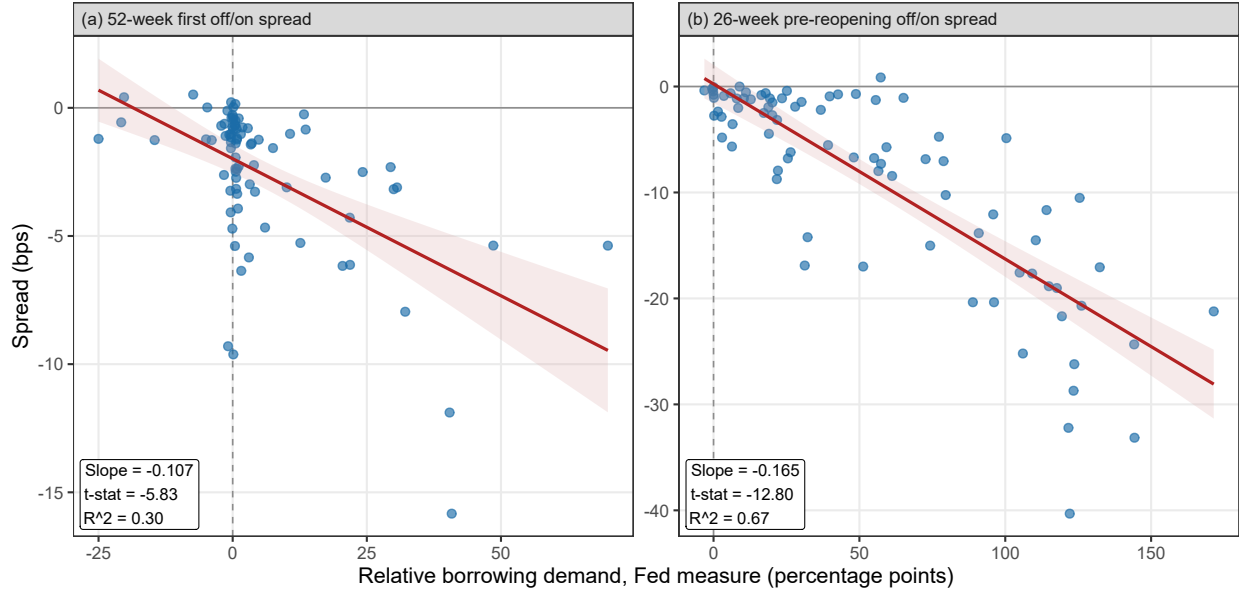


Figure 3: On-the-run discounts and relative borrowing demand. This figure plots the two on-the-run discounts of Figure 1 against the corresponding borrowing-demand spread in the Federal Reserve’s securities-lending operations, measured as borrowing demand for the seasoned leg minus borrowing demand for the newly issued leg, in percentage points. Panel (a) shows the 52-week first off/on pair, with one observation per auction cycle. Panel (b) shows the 26-week pre-reopening off/on pair, with one observation per reopening window. Yield spreads and borrowing-demand spreads are averaged over the same days. The lines are OLS fits with 95% confidence bands; the boxes report the slope, its t -statistic, and the R^2 . The sample covers auction cycles and reopening windows with Federal Reserve lending data, 2019–2025.

We estimate:

$$\Delta Spread_{p,t} = \alpha_p + \theta \Delta BD Spread_{p,t} + \beta \Delta AS Spread_{p,t} + \delta' \Delta Z_{p,t} + \gamma' X_t + \varepsilon_{p,t}, \quad (3)$$

where p denotes the 52-week first off/on pair or the 26-week pre-reopening pair. The dependent variable is the change in the pair's OIS-adjusted yield spread, $\Delta BD Spread_{p,t}$ is the corresponding change in relative borrowing demand expressed as a ratio rather than in percentage points, and $\Delta AS Spread_{p,t}$ is the change in the relative available-supply control. In the repo-specialness specifications, $\Delta BD Spread_{p,t}$ is replaced by the change in the pair's repo-specialness spread. The vector $\Delta Z_{p,t}$ contains changes in pair-specific auction-demand and CDS spreads. The market controls in X_t are changes in the bill-to-GDP ratio, MOVE, and the one-year-minus-one-month OIS slope, together with the level of the one-month OIS rate.¹⁴

Panel A of Table 5 reports the results for the 52-week first off/on spread. In Columns (1) and (2), a unit increase in the Federal Reserve borrowing-demand spread lowers the yield spread by 7.72 basis points without controls and 7.65 basis points with controls. The corresponding Markit coefficients in Columns (3) and (4) are -1.16 and -1.00 , while a one-basis-point increase in relative repo specialness lowers the yield spread by 0.03 and 0.02 basis points in Columns (5) and (6). All three measures therefore show that greater borrowing pressure on the seasoned bill is associated with a larger on-the-run discount. The coefficients remain statistically significant after adding the controls.

Panel B reports the corresponding results for the 26-week pre-reopening spread. The

¹⁴An alternative would be to measure Treasury richness as in Fleckenstein and Longstaff (2024) by directly adjusting for differences in CDS premiums. Because the OIS and OIS-plus-CDS adjustments are highly correlated and the CDS control is not statistically significant, we use the OIS adjustment in the main specification.

Table 5: Changes in the on-the-run discounts and relative borrowing pressure. This table reports window-to-window regressions of Equation (3) for the two on-the-run discounts of Figure 1. The dependent variable in Panel A is the change in the OIS-adjusted 52-week first off/on spread, $\Delta Spread_{52w,t}$, one observation per auction cycle, and in Panel B the change in the 26-week pre-reopening off/on spread, $\Delta Spread_{26w,t}$, one observation per reopening window, both in basis points. Each relative measure is constructed as the value for the seasoned leg minus the value for the newly issued leg, averaged over the corresponding window. $\Delta BD^{Fed} Spread_t$ uses par submitted relative to par available in the Federal Reserve’s securities-lending operations and $\Delta BD^{Mkit} Spread_t$ uses Markit’s short-loan quantity relative to actively lendable quantity, both expressed as ratios rather than percentages. $\Delta Specialness Spread_t$ uses overnight repo specialness in the DTCC data, in basis points, so positive values indicate that the seasoned bill is more special. $\Delta AS Spread_t$ is the change in the log available supply of the seasoned leg minus that of the newly issued leg, where available supply is the par amount awarded to primary dealers at the bill’s initial auction. The even columns add this control, the pair’s auction-demand and CDS spreads, and the market controls specified in Equation (3); the market controls are not reported. The numbers in parentheses are t -statistics. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

<i>Panel A. 52-week first off/on spread, $\Delta Spread_{52w,t}$</i>						
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta BD^{Fed} Spread_t$	-7.72*** (-6.44)	-7.65*** (-5.97)				
$\Delta BD^{Mkit} Spread_t$			-1.16*** (-3.46)	-1.00*** (-3.05)		
$\Delta Specialness Spread_t$					-0.03*** (-6.97)	-0.02*** (-5.45)
$\Delta AS Spread_t$		0.83* (1.79)		1.58*** (5.75)		1.04*** (3.41)
$\Delta AuctionDemand Spread_t$		-0.04 (-1.19)		0.02 (1.06)		0.01 (0.69)
$\Delta CDS Spread_t$		-0.25 (-1.37)		0.003 (0.02)		-0.04 (-0.30)
Market controls	No	Yes	No	Yes	No	Yes
Sample	2019–2025	2019–2025	2010–2025	2010–2025	2011–2025	2011–2025
Observations	81	80	203	200	186	186
Adjusted R ²	0.34	0.39	0.05	0.19	0.20	0.26
<i>Panel B. 26-week pre-reopening off/on spread, $\Delta Spread_{26w,t}$</i>						
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta BD^{Fed} Spread_t$	-6.83*** (-5.09)	-7.52*** (-5.11)				
$\Delta BD^{Mkit} Spread_t$			-1.24** (-2.14)	-1.21** (-2.02)		
$\Delta Specialness Spread_t$					-0.03*** (-11.49)	-0.04*** (-13.17)
$\Delta AS Spread_t$		1.54 (0.85)		0.68 (0.77)		0.47 (0.69)
$\Delta AuctionDemand Spread_t$		0.06 (0.48)		-0.04 (-0.58)		0.06 (1.27)
$\Delta CDS Spread_t$		1.28 (1.51)		1.01 (1.64)		2.48*** (5.20)
Market controls	No	Yes	No	Yes	No	Yes
Sample	2019–2025	2019–2025	2010–2025	2010–2025	2011–2025	2011–2025
Observations	81	77	201	197	186	183
Adjusted R ²	0.24	0.22	0.02	0.004	0.41	0.49

Federal Reserve coefficients are -6.83 and -7.52 , the Markit coefficients are -1.24 and -1.21 , and the repo-specialness coefficients are -0.03 and -0.04 . Relative Federal Reserve borrowing demand alone explains 24% of the variation in yield-spread changes, while relative specialness alone explains 41%. Because Markit measures the stock of outstanding loans whereas the Federal Reserve measure records bids at a backstop facility, the coefficient magnitudes are not directly comparable. Table IA.9 in the Internet Appendix shows that the Markit and repo-specialness coefficients remain negative and statistically significant on the cycles and windows with Federal Reserve lending data.

Finally, Table IA.10 in the Internet Appendix examines the third prediction using adjacent pairs of seasoned bills and finds negative coefficients for all five pairs with both borrowing-demand measures, statistically significant for all five using the Federal Reserve measure and for the first four using Markit, showing that the relationship between borrowing demand and relative prices extends beyond the on-the-run label.

5 Implications of the On-the-run Discount

In this section, we draw out two implications of the on-the-run discount. Each concerns an established regularity in money markets that the reopening cycle interrupts. First, the convenience premium of US Treasuries increases as short-term Treasuries approach their maturity date. We document a sharp discontinuity in this pattern when a new 26-week bill is issued. Second, money market mutual funds (MMFs) prefer Treasury bills with short maturities and are widely regarded as buy-and-hold investors. Both descriptions hold for bills with less than six months to maturity. However, for 52-week bills, MMFs tilt toward longer maturities and reduce their positions as the reopening approaches.

5.1 Convenience Premiums of Near-Money Assets

The literature measures Treasury convenience premiums by comparing the yield of a Treasury security with a “risk-free” reference rate that carries no convenience services. Examples of reference rates include OIS rates referencing the effective federal funds rate (e.g., Klingler and Sundaresan, 2023; He et al., 2020), OIS rates referencing SOFR combined with a CDS contract on the US Treasury (Fleckenstein and Longstaff, 2024), and yields from a Treasury curve fitted on longer maturities (e.g., Greenwood et al., 2015).¹⁵ While reference rates vary across studies, it is common to measure the convenience premium using on-the-run Treasuries.

A common view is that convenience premiums rise as maturity shortens, because shorter-term bills are closer substitutes for money (Krishnamurthy and Vissing-Jorgensen, 2012, Greenwood et al., 2015, Nagel, 2016, Fleckenstein and Longstaff, 2024). While we confirm this overall pattern, we highlight that the on-the-run discount introduces a sharp discontinuity to this pattern.

Figure 4 plots the term structure of OIS–Treasury spreads. Panel (a) shows the spread for original 52-week bills against bill age. As expected, the spread widens as the bill ages and approaches its 26-week reopening. However, richness drops sharply at the reopening and starts rebuilding as the bill 26-week bill ages. Panel (b) repeats the exercise for 2-year and 10-year notes with less than one year to maturity. The richness of these securities also rises as maturity shortens. However, they are not reopened when they cross 26 weeks to maturity, and therefore their richness does not drop at 26 weeks.

¹⁵Other reference rates include the risk-free rate implied by put-call parity in index options (e.g., van Binsbergen et al., 2022), term repo rates (e.g., Nagel, 2016), the yield of highly rated commercial paper (e.g., Krishnamurthy and Vissing-Jorgensen, 2012), and the yield of Federal Home Loan Bank discount notes (e.g., Nagel, 2016; Klingler and Sundaresan, 2023). We focus on OIS rates and extrapolated Treasury yields because we require a reference rate available in granular maturity buckets between 52 and 22 weeks.

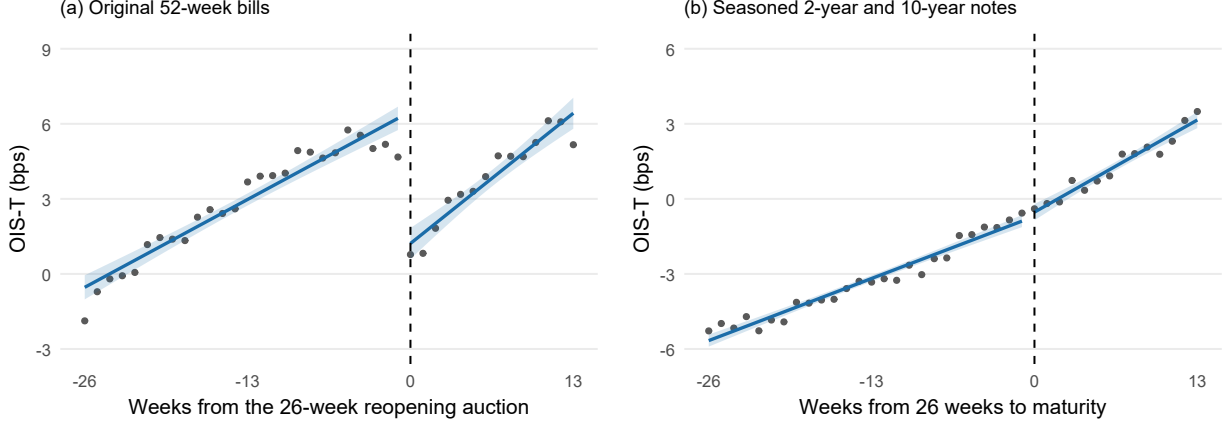


Figure 4: Average OIS–Treasury spread around the 26-week reopening: bills versus notes. Each dot shows the average OIS–Treasury spread, measured as the maturity-matched OIS rate minus the security’s yield, in basis points; higher values indicate that the security is richer relative to OIS. Panel (a) plots original 52-week bills by the number of weeks between the observation date and the bill’s 26-week reopening auction; the vertical dashed line marks the auction. Panel (b) plots seasoned 2-year and 10-year notes in the last year of their life, pooled across the two tenors and aligned at 26 weeks to maturity, the point at which a 52-week bill is reopened; these securities face no reopening. The solid lines show separate weighted linear fits on each side of the dashed line, with shaded 95% confidence intervals. The sample period is 2010–2025.

Table 6 quantifies the lifecycle pattern for three different measures of the convenience premium. For each measure, we estimate the following regression:

$$CY_{i,t} = \beta_1 \mathbb{1}(\text{Reopened})_{i,t} + \beta_2 \text{Distance}_{i,t} + \beta_3 \text{Distance}_{i,t} \times \mathbb{1}(\text{Reopened})_{i,t} + FE_t + \epsilon_{i,t}, \quad (4)$$

where $CY_{i,t}$ is the convenience premium of bill i on date t , $\mathbb{1}(\text{Reopened})_{i,t}$ is an indicator variable that equals one from the day of the bill’s 26-week reopening auction, $\text{Distance}_{i,t}$ is the number of days between date t and the reopening auction (negative before the auction), and FE_t are date fixed effects that distinguish the observed pattern from any seasonal effects.

As shown in Table 6, all three measures drop sharply after the reopening. This drop ranges from 5.8 to 6.3 basis points. Given that the average OIS-Treasury spread for 26-week on-the-run bills is around 0.5 basis points, this discontinuity highlights that the magnitude of the convenience premium is strongly affected by when in a bill’s life cycle the premium is measured. In addition, the table confirms that convenience premiums increase with shorter times to maturity.

5.2 Money Market Funds Prefer On-the-Run Bills

In this section, we examine whether MMFs internalize the on-the-run discount. Most investors hold Treasury bills for their safety and liquidity and accept small yield spreads for the convenience of holding safe assets. They are therefore not sensitive to small yield differentials between off- and on-the-run bills. MMFs are the exception. They are large investors in the market for Treasury bills (e.g., Doerr et al., 2023) and compete aggressively for small yield differentials (Kacperczyk and Schnabl, 2013; La Spada, 2018). MMFs cannot take short positions, but they can reduce their positions in expensive off-the-run bills and hold more of the cheaper on-the-run bills.

To investigate this possibility, we obtain MMF holdings of Treasury bills from Crane data. We focus on Government and Treasury MMFs because other fund types hold virtually no Treasury bills. Panel (a) of Figure 5 shows aggregate average MMF holdings of 52-week bills by remaining maturity, measured as a fraction of the amount outstanding.¹⁶ We examine the short end (less than six months to maturity) separately from the longer end (7 to 12 months to maturity). For the longer end, MMF holdings decrease as the bill ages. MMFs thus hold a larger share of the cheaper on-the-run bills and reduce their positions as the bill approaches

¹⁶Figure IA.4 in the Internet Appendix shows that this pattern is virtually identical when we include all Treasury bills.

Table 6: Convenience premiums over the bill’s life. This table reports daily panel regressions of three convenience-premium measures for original 52-week Treasury bills on the bill’s distance to its 26-week reopening auction, an indicator that equals one from the auction day, and their interaction. Each measure is a reference rate at the bill’s remaining maturity minus the bill’s yield, in basis points, so higher values mean that the bill is richer relative to the reference rate. Column (1) uses the maturity-matched OIS rate referencing the effective federal funds rate. Column (2) uses the synthetic risk-free rate of Fleckenstein and Longstaff (2024), SOFR OIS rates combined with the U.S. sovereign CDS premium, from a replication of their measure. Column (3) uses the fitted Treasury curve of Gürkaynak et al. (2007) evaluated at the bill’s remaining maturity. $Distance_{i,t}$ is the number of days between date t and bill i ’s 26-week reopening auction, negative before it, and $\mathbb{1}(\text{Reopened})_{i,t}$ equals one from the auction day, so its coefficient is the jump when the reopening supply is priced; an additional indicator for the settlement, three days later, yields no further jump (untabulated). The window runs from each bill’s 52-week auction to four weeks after its reopening settlement, the window of Table 4. All specifications include date fixed effects. The numbers in parentheses are t -statistics based on standard errors clustered by date and CUSIP. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. The regression equation is:

$$CY_{i,t} = \beta_1 \mathbb{1}(\text{Reopened})_{i,t} + \beta_2 Distance_{i,t} + \beta_3 Distance_{i,t} \times \mathbb{1}(\text{Reopened})_{i,t} + FE_t + \epsilon_{i,t}.$$

	OIS-T (1)	FL (2)	GSW (3)
$\mathbb{1}(\text{Reopened})_{i,t}$	−6.299*** (−10.500)	−6.227*** (−10.321)	−5.868*** (−9.812)
$Distance_{i,t}$ (days)	0.039*** (17.626)	0.030*** (13.207)	0.050*** (14.658)
$Distance_{i,t} \times \mathbb{1}(\text{Reopened})_{i,t}$	0.069*** (9.957)	0.072*** (7.322)	0.087*** (6.925)
Date FE	Yes	Yes	Yes
Sample	2010–2025	2010–2025	2010–2025
Bills	203	203	203
Observations	29,771	29,289	29,777
R ²	0.850	0.948	0.625
Adjusted R ²	0.827	0.940	0.567

its reopening. This tilt toward bills with longer maturities is especially surprising given that regulatory rules generally push MMFs toward shorter-maturity instruments. By contrast,

for bills with six or fewer months to maturity, holdings increase as maturity shortens.

Panel (b) shows that MMF holdings of Treasury notes in their final year display no such reversal. The MMF share rises monotonically as maturity shortens. Hence, MMFs' tilt toward longer-term bills is specific to securities that are about to be reopened. Hence, Figure 5 suggests that MMFs are aware of the on-the-run discount and repond by holding more of the cheaper on-the-run securities. However, given that MMFs are relatively small investors in Treasury bills with longer maturities, it is plausible that their investment behavior is not enough to correct the on-the-run discount.

While Figure 5 is suggestive evidence that MMFs are aware of the on-the-run discount, we next examine the association between their Treasury holdings and the on-the-run discount more closely. We construct the volume-weighted aggregate duration of MMF holdings of Treasury bills with more than 252 days to maturity ($Duration_t$). We then run linear regressions of the following form:

$$Duration_t = \alpha + \beta Spread_{0,t}^4 + \gamma' Controls_t + \epsilon_t. \quad (5)$$

The main dependent variable in Equation (5) is the aggregate maturity of MMF Treasury bill portfolios. We run the regressions in both levels and first differences. The level specifications include year fixed effects, and the controlled specifications include aggregate MMF weighted average life and flows.

Table 7 shows the results of our analysis. The coefficient on $Spread_{0,t}^4$ is negative and statistically significant across specifications. Since more negative values of $Spread_{0,t}^4$ correspond to a deeper on-the-run discount, the estimates imply that deeper discounts are associated with longer MMF bill portfolios and a larger tilt toward bills with a longer time to maturity.

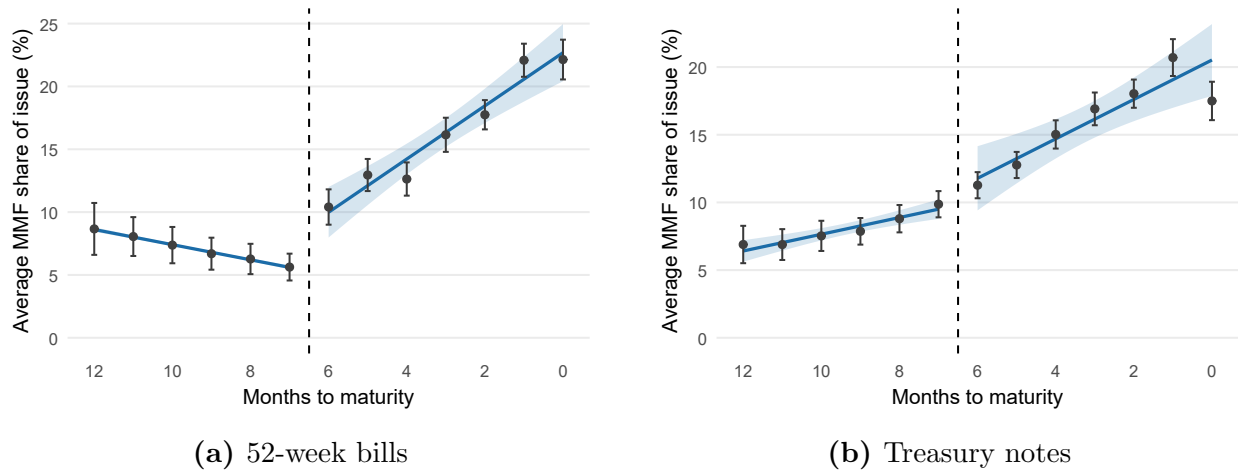


Figure 5: Money market fund holdings by remaining maturity: bills versus notes. This figure plots the average fraction of each issue held by money market funds, in percent of the amount outstanding, by remaining maturity in months, using month-end Crane holdings matched to Treasury issuance data. Panel (a) covers securities issued as 52-week Treasury bills, followed through their reopenings (178 bills, October 2011 to July 2025). Panel (b) covers Treasury notes in the last year of their life (758 securities, same period). Each dot is the average across issue-month observations in that months-to-maturity bin; whiskers denote 95% confidence intervals. The solid lines show separate weighted linear fits for the 7-to-12-month and 0-to-6-month segments, with shaded 95% confidence intervals, and the vertical dashed line marks 26 weeks to maturity, the point at which a 52-week bill is reopened. For bills, the share falls as the security ages toward its reopening and rises afterwards; for notes, the share rises monotonically as maturity shortens.

The result is robust in both the level specifications with year fixed effects and the first-difference specifications. This evidence is consistent with money market funds responding to cheap on-the-run bills by buying longer-maturity bills and reducing their holdings of more expensive shorter-maturity bills.

Table 7: On-the-run discounts and money market fund portfolio maturity. This table reports monthly regressions of the maturity profile of money market fund Treasury bill portfolios on the 52-week bill on-the-run discount. The dependent variable, Duration, is the principal-weighted remaining maturity, in months, of MMF Treasury bill holdings with more than 252 days remaining to maturity. The key explanatory variable, $Spread_{0,t}^4$, is the OIS-adjusted spread between the fourth off-the-run 52-week bill and the current on-the-run bill; more negative values indicate a deeper on-the-run discount. Columns (1)–(2) use levels and include year fixed effects, while Columns (3)–(4) use first differences. Columns (2) and (4) include aggregate MMF weighted average life and flows as controls. The sample is monthly and determined by the overlap of Crane holdings and 52-week spread data. The numbers in parentheses are t -statistics. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. The regression equation is:

$$Duration_t = \alpha + \beta Spread_{0,t}^4 + \gamma' Controls_t + \epsilon_t.$$

	Duration			
	(1)	(2)	(3)	(4)
$Spread_{0,t}^4$	−0.03** (−2.45)	−0.03** (−2.51)	−0.04** (−2.59)	−0.04** (−2.42)
WAL		0.004 (0.26)		0.04* (1.70)
Flow		0.61 (0.51)		−1.09 (−0.80)
Constant			−0.01 (−0.15)	−0.01 (−0.17)
Specification	Levels	Levels	Changes	Changes
Year FE	Yes	Yes	No	No
Observations	163	160	160	157
Adjusted R ²	−0.01	−0.02	0.03	0.04

6 Conclusion

We show that the most recently issued US Treasury bill trades at a discount relative to older Treasury bills of comparable maturity. This on-the-run discount for Treasury bills contrasts with coupon-bearing on-the-run Treasuries, which trade at a premium relative to comparable seasoned securities. We link the on-the-run discount to the Treasury bill reopening cycle and short-selling activity. As a bill approaches its reopening, short-selling

demand and repo specialness increase for that security, and the bill becomes richer in the cash market. Both patterns reverse after the reopening delivers additional securities of the same CUSIP. Viewed from the other side of the trade, the discount reflects a reopening premium, with seasoned bills commanding higher prices.

The reopening cycle has three broader implications. First, it interrupts the term structure of convenience premiums. The convenience premium of a 52-week bill rises as the bill ages, drops by roughly 6 basis points at the reopening auction, and then builds anew. Hence, the measured convenience of a near-money asset depends on where in the reopening cycle the security sits. Second, MMFs respond to the discount by holding more of the cheaper on-the-run bills and reducing their positions in seasoned bills as the reopening approaches. Third, for Treasury debt management, spreading bill issuance across a finer set of maturities may reduce the relative underpricing of newly issued bills and lower Treasury borrowing costs. The recent introduction of 8-week and 17-week bills is a step in this direction, and an additional maturity between 26 and 52 weeks could further lower borrowing costs.

References

- Adrian, T., M. J. Fleming, and E. Vogt (2017). An index of Treasury market liquidity: 1991-2017. Working paper, Federal Reserve Bank of New York.
- Amihud, Y. and H. Mendelson (1991). Liquidity, maturity, and the yields on us treasury securities. *The Journal of Finance* 46(4), 1411–1425.
- Augustin, P., M. Chernov, L. Schmid, and D. Song (2021). Benchmark interest rates when the government is risky. *Journal of Financial Economics* 140(1), 74–100.

- Barclay, M. J., T. Hendershott, and K. Kotz (2006). Automation versus intermediation: Evidence from treasuries going off the run. *The Journal of Finance* 61(5), 2395–2414.
- Beetsma, R., M. Giuliodori, F. De Jong, and D. Widiyanto (2016). Price effects of sovereign debt auctions in the euro-zone: The role of the crisis. *Journal of Financial Intermediation* 25, 30–53.
- Bikhchandani, S., P. L. Edsparr, and C.-f. Huang (2000). The treasury bill auction and the when-issued market: Some evidence. UCLA Anderson Graduate School of Management Working Paper.
- Brain, D., M. De Pooter, D. Dobrev, M. Fleming, P. Johansson, F. Keane, M. Puglia, T. Rodrigues, and O. Shachar (2018). Breaking down TRACE volumes further. *Liberty Street Economics Blog*.
- Cammack, E. B. (1991). Evidence on bidding strategies and the information in treasury bill auctions. *Journal of Political Economy* 99(1), 100–130.
- D’Amico, S., R. Fan, and Y. Kitsul (2018). The scarcity value of treasury collateral: Repo-market effects of security-specific supply and demand factors. *Journal of Financial and Quantitative Analysis* 53(5), 2103–2129.
- D’Amico, S. and N. A. Pancost (2022). Special repo rates and the cross-section of bond prices: The role of the special collateral risk premium. *Review of Finance* 26(1), 117–162.
- Dittmar, R. F., A. Hsu, G. Roussellet, and P. Simasek (2026). Default risk and the pricing of us sovereign bonds. *The Journal of Finance* 81(2), 829–869.
- Doerr, S., E. Eren, and S. Malamud (2023). Money market funds and the pricing of near-money assets. Working paper, Swiss Finance Institute.

- Du, W., J. Im, and J. Schreger (2018). The U.S. Treasury Premium. *Journal of International Economics* 112, 167–181.
- Du, W., A. Tepper, and A. Verdelhan (2018). Deviations from covered interest rate parity. *The Journal of Finance* 73(3), 915–957.
- Duffee, G. R. (1996). Idiosyncratic variation of treasury bill yields. *The Journal of Finance* 51(2), 527–551.
- Duffie, D. (1996). Special repo rates. *The Journal of Finance* 51(2), 493–526.
- Duffie, D. (2017). Financial Regulatory Reform After the Crisis: An Assessment. *Management Science* 64(10), 4835–4857.
- Duffie, D. (2020). Still the world’s safe haven? redesigning the U.S. Treasury market after the COVID-19 crisis. Working paper, Brookings Institution.
- Fleckenstein, M. and F. A. Longstaff (2024). Treasury richness. *The Journal of Finance* 79(4), 2797–2844.
- Fleming, M., G. Nguyen, and J. Rosenberg (2024). How do treasury dealers manage their positions? *Journal of Financial Economics* 158, 103885.
- Fleming, M. J. (2002). Are larger treasury issues more liquid? evidence from bill reopenings. *Journal of Money, Credit and Banking*, 707–735.
- Fleming, M. J. (2003). Measuring treasury market liquidity. *Federal Reserve Bank of New York Staff Report* (133).
- Fleming, M. J., W. Liu, and G. Nguyen (2026). Intraday price pressure and order flow around us treasury auctions. *Federal Reserve Bank of New York Staff Report* (1188).

- Fontaine, J. S. and R. Garcia (2012). Bond liquidity premia. *The Review of Financial Studies* 25(4), 1207–1254.
- Goldreich, D. (2007). Underpricing in discriminatory and uniform-price treasury auctions. *Journal of Financial and Quantitative Analysis* 42(2), 443–466.
- Goldreich, D., B. Hanke, and P. Nath (2005). The price of future liquidity: Time-varying liquidity in the us treasury market. *Review of Finance* 9(1), 1–32.
- Goyenko, R., A. Subrahmanyam, and A. Ukhov (2011). The term structure of bond market liquidity and its implications for expected bond returns. *Journal of Financial and Quantitative Analysis* 46(1), 111–139.
- Greenwood, R., S. G. Hanson, and J. C. Stein (2015). A comparative-advantage approach to government debt maturity. *The Journal of Finance* 70(4), 1683–1722.
- Gürkaynak, R. S., B. Sack, and J. H. Wright (2007). The US Treasury yield curve: 1961 to the present. *Journal of Monetary Economics* 54(8), 2291–2304.
- He, Z., S. Nagel, and Z. Song (2020). Treasury inconvenience yields during the covid-19 crisis. Working paper, National Bureau of Economic Research.
- Herb, P. (2018). Are US Treasury auctions twice underpriced? Working paper, Northern Arizona University.
- Ho, T. and H. R. Stoll (1981). Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics* 9(1), 47–73.
- Hu, G. X., J. Pan, and J. Wang (2013). Noise as information for illiquidity. *The Journal of Finance* 68(6), 2341–2382.

- Jiang, Z., R. J. Richmond, and T. Zhang (2025). Convenience lost. Working paper, National Bureau of Economic Research.
- Jordan, B. D. and S. D. Jordan (1997). Special repo rates: An empirical analysis. *The Journal of Finance* 52(5), 2051–2072.
- Kacperczyk, M. and P. Schnabl (2013). How safe are money market funds? *Quarterly Journal of Economics* 128(3), 1073–1122.
- Klingler, S. and D. Lando (2018). Safe haven CDS premiums. *The Review of Financial Studies* 31(5), 1855–1895.
- Klingler, S. and S. Sundaresan (2023). Diminishing treasury convenience premiums: Effects of dealers’ excess demand and balance sheet constraints. *Journal of Monetary Economics* 135, 55–69.
- Klingler, S. and O. Syrstad (2021). Life after libor. *Journal of Financial Economics* 141(2), 783–801.
- Krishnamurthy, A. (2002). The bond/old-bond spread. *Journal of Financial Economics* 66(2-3), 463–506.
- Krishnamurthy, A. and A. Vissing-Jorgensen (2011). The effects of quantitative easing on interest rates: Channels and implications for policy. Technical report, National Bureau of Economic Research.
- Krishnamurthy, A. and A. Vissing-Jorgensen (2012). The aggregate demand for treasury debt. *Journal of Political Economy* 120(2), 233–267.

- La Spada, G. (2018). Competition, reach for yield, and money market funds. *Journal of Financial Economics* 129(1), 87–110.
- Longstaff, F. A. (2004). The flight-to-liquidity premium in US Treasury bond prices. *Journal of Business* 77(3), 511–526.
- Lou, D., H. Yan, and J. Zhang (2013). Anticipated and repeated shocks in liquid markets. *The Review of Financial Studies* 26(8), 1891–1912.
- Lowenstein, R. (2001). *When genius failed: The rise and fall of Long-Term Capital Management*. Random House trade paperbacks.
- Musto, D., G. Nini, and K. Schwarz (2018). Notes on bonds: Illiquidity feedback during the financial crisis. *The Review of Financial Studies* 31(8), 2983–3018.
- Nagel, S. (2016). The liquidity premium of near-money assets. *Quarterly Journal of Economics* (December), 1927–1971.
- Pasquariello, P. and C. Vega (2009). The on-the-run liquidity phenomenon. *Journal of Financial Economics* 92(1), 1–24.
- Pedersen, L. H. (2019). *Efficiently inefficient: How smart money invests and market prices are determined*. Princeton University Press.
- Ray, W., M. Droste, and Y. Gorodnichenko (2024). Unbundling quantitative easing: Taking a cue from treasury auctions. *Journal of Political Economy* 132(9), 3115–3172.
- Sigaux, J.-D. (2024). Trading ahead of treasury auctions. *Journal of Banking & Finance* 158, 107032.

- Spindt, P. A. and R. W. Stolz (1992). Are u.s. treasury bills underpriced in the primary market? *Journal of Banking & Finance* 16(5), 891–908.
- Sundaresan, S. (2009). *Fixed income markets and their derivatives*. Academic Press.
- van Binsbergen, J. H., W. F. Diamond, and M. Grotteria (2022). Risk-free interest rates. *Journal of Financial Economics* 143(1), 1–29.
- Vayanos, D. and P.-O. Weill (2008). A search-based theory of the on-the-run phenomenon. *The Journal of Finance* 63(3), 1361–1398.
- Veronesi, P. (2016). *Handbook of fixed-income securities*. John Wiley & Sons.
- Warga, A. (1992). Bond returns, liquidity, and missing data. *Journal of Financial and Quantitative Analysis* 27(4), 605–617.

Table 8: Variable description. This table defines the variables used in our empirical analysis and shows the datasources.

Variable	Description	Source
Treasury yields	Daily Treasury prices are used to compute yields, remaining maturity	CRSP Treasury
Benchmark rates	OIS rates are interpolated to each security’s remaining maturity and used as the main benchmark rate. GSW zero-coupon yields provide an alternative maturity adjustment.	Bloomberg, Fed
CDS adjustment	U.S. sovereign CDS spreads are interpolated across maturities and added to OIS-based benchmark adjustments in robustness checks.	Markit
On-the-run spreads	Yield spreads compare the first, second, and older off-the-run securities with the current on-the-run security of the same original tenor. The baseline spread adjusts for the maturity mismatch using matched-maturity OIS rates.	Author calculations
Pre-reopening off/on spread	OIS-adjusted yield of the 52-week bill approaching its 26-week reopening minus the yield of the on-the-run 26-week bill, averaged within each window between reopening auctions; days on which the on-the-run bill is a reopened 52-week security are excluded.	CRSP Treasury
Repo rates and specialness	Security-level overnight repo rates are volume-weighted repo rates. Repo specialness is the benchmark overnight repo rate minus the security-specific repo rate.	DTCC
Overnight repo benchmark	The overnight repo benchmark is SOFR when available. For the pre-SOFR period, synthetic SOFR is constructed as the volume-weighted average of the general-collateral repo rate and the BNY Tri-Party Repo Index.	DTCC and BNY Tri-Party Repo Index
Securities-lending borrowing demand	Results of the Federal Reserve’s securities-lending operations report, for each security and operation date, the par amount submitted by primary dealers and the par amount available for lending. Borrowing demand is par submitted relative to par available, in percent. For the auction-cycle measures, days without bids count as zero once the security is available for lending.	Federal Reserve Bank of New York
Securities-lending short loans	Daily security-level short-loan quantity and actively lendable quantity; the short-loan share is their ratio.	Markit Securities Finance
Bid-ask spreads	Bid-ask spreads are measured from CRSP daily quotes and GovPX interdealer quotes. For bills, GovPX discount-rate quotes are converted to prices before computing price bid-ask spreads.	CRSP and GovPX
Auction results and allocations	Auction controls include offering characteristics, accepted amounts, bid-to-cover ratios, auction yields, cash-management-bill indicators, and investor-class allotments.	TreasuryDirect
Reopening event time	Distance is the number of calendar days from the settlement of the bill’s 26-week reopening, negative before it; the reopened indicator equals one from the settlement day.	TreasuryDirect
Auction demand shock	The auction demand shock is the intraday yield change around auction results, measured using GovPX quotes shortly before the auction close and shortly after the Bloomberg auction-result timestamp.	GovPX and Bloomberg
Bill/GDP	Treasury supply controls scale bill or non-bill Treasury amounts by aggregate GDP and enter the regressions as changes at the auction-cycle frequency.	Klingler and Syrstad (2021)
Market conditions	Market-wide controls include the MOVE index, the VIX, the Treasury noise measure, the one-month OIS rate, and the money-market slope, measured as the one-year minus the one-month OIS rate.	Bloomberg, CBOE, CRSP
Money market fund holdings	Monthly Treasury holdings by money market funds are aggregated to the CUSIP level and scaled by the issue’s outstanding amount to measure the share held by MMFs.	Crane Data

Internet Appendix for:

The On-the-Run Discount

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September 17, 2026

This document contains additional descriptions, figures, tables, and robustness checks that supplement the paper titled “The On-the-Run Discount.”

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IA.1 Additional Information on Data Construction

This section provides additional details on the construction of the main variables used in the paper.

IA.1.1 Treasury Bill Yield Calculation

Treasury bills are quoted on a bank-discount basis. We convert bill discount rates into prices and coupon-equivalent yields following the Treasury bill calculation methodology. Let d denote the discount rate, r the number of days to maturity, y the number of days in the year, and P the price per \$100 of face value. The price implied by a discount-rate quote is

$$P = 100 \left(1 - \frac{dr}{360} \right).$$

For bills with no more than one half-year to maturity, the coupon-equivalent yield, or investment rate, is

$$i = \left(\frac{100 - P}{P} \right) \left(\frac{y}{r} \right).$$

For bills with more than one half-year to maturity, the yield solves the Treasury's semiannual-compounding equation,

$$P \left[1 + \left(r - \frac{y}{2} \right) \frac{i}{y} \right] \left(1 + \frac{i}{2} \right) = 100,$$

or equivalently

$$\left(\frac{r}{2y} - 0.25 \right) i^2 + \frac{r}{y} i + \frac{P - 100}{P} = 0.$$

We use the economically relevant root of this quadratic equation and keep intermediate calculations at full precision, applying rounding only to displayed output.

IA.1.2 Coupon-Security Curve Adjustment

For coupon-bearing Treasury securities, we use the Gürkaynak et al. (2007) zero-coupon yield curve to construct a curve-implied price that accounts for both maturity and coupon differences. For each security i on date t , with cash flows $\{CF_{i,\tau}\}$ at payment dates $\tau > t$, we compute

$$P_{i,t}^{\text{GSW}} = \sum_{\tau > t} CF_{i,\tau} D_t^{\text{GSW}}(\tau),$$

where $D_t^{\text{GSW}}(\tau)$ is the discount factor implied by the GSW zero-coupon curve. Using the same yield convention as for the observed market price $P_{i,t}$, we convert $P_{i,t}^{\text{GSW}}$ into a curve-implied yield $y_{i,t}^{\text{GSW}}$ and define the curve-based yield deviation as

$$m_{i,t}^{\text{GSW}} = y_{i,t} - y_{i,t}^{\text{GSW}}.$$

IA.1.3 External Treasury-Market Measures

The empirical analysis also uses Treasury-market controls and comparison measures, including the note/bill spread, the Treasury noise measure, and auction demand proxies.

Note-bill spread. We construct the note-bill spread from CRSP Treasury prices following the spirit of Amihud and Mendelson (1991). For each date, we compute dirty midpoint prices as the midpoint of the bid and ask prices plus accrued interest. For bills, we convert the dirty price into an annualized yield, $100[(100/P)^{365/r} - 1]$, where P is the dirty price per \$100 face

value and r is remaining maturity in days. For coupon notes, we select the note closest to the target remaining maturity and compute a comparable standardized yield from the dirty price and the semiannual coupon component. We then linearly interpolate bill yields to the exact remaining maturity of the selected note using the two bracketing bills on the same date. The note-bill spread is the note yield minus this interpolated bill yield, expressed in basis points. In the main correlation table, we use the version based on notes with remaining maturities between 120 and 180 days and target maturity closest to 150 days.

Noise measure. We replicate the Treasury noise measure of Hu et al. (2013) using CRSP Treasury prices and fitted Nelson–Siegel–Svensson yields. For each Treasury note or bond with one to ten years remaining maturity, we compute the pricing error as the absolute difference between the observed midpoint yield and the fitted curve-implied yield. We exclude observations with invalid quotes or nonpositive yields and remove large daily outliers by dropping securities whose pricing error exceeds four times the cross-sectional standard deviation on that date. The daily noise measure is the root mean squared pricing error across the remaining securities, multiplied by 100 so that the series is expressed in basis points. When constructing the regression and correlation datasets, we merge this replicated series with the available benchmark noise series and use the benchmark series when available.

Auction demand proxies. Our main auction-demand proxy is an intraday yield reaction around Treasury auction results, following the logic of Ray et al. (2024). We use GovPX intraday quotes matched to the auctioned CUSIP on the auction date. The pre-auction yield is the last observed midpoint yield at least ten minutes before the competitive auction close. The post-auction yield is the first observed midpoint yield at least ten minutes after the Bloomberg auction-result timestamp. The demand shock is the post-auction midpoint yield

minus the pre-auction midpoint yield, expressed in basis points, so positive values correspond to yields rising around the auction result. We also retain standard auction-demand controls from TreasuryDirect, including the bid-to-cover ratio and the auction tail, measured as the high yield minus the pre-auction when-issued yield. The latter is computed from GovPX midpoint yields averaged from 180 minutes to one minute before the auction close.

IA.2 Additional Descriptive Statistics

This section reports additional descriptive statistics that support the main sample construction and comparisons. Table IA.1 reports level correlations between the on-the-run discount and other Treasury-market measures. Table IA.2 documents the two borrowing-demand measures, with their numerators and denominators, for original 52-week bills before their 26-week reopening, and the borrowing-demand spread of the two bill pairs, whose window-to-window changes are the key regressors in Section 4.3, and of the 2-year and 10-year note pairs. Table IA.3 reports CRSP and GovPX bid-ask spreads for the four bills that form the two pairs. Table IA.4 reports additional summary statistics for coupon securities and their relation to adjacent-issue spreads. Figure IA.1 plots the two bill on-the-run discounts of Figure 1 together with the level of the one-month OIS rate.

Table IA.1: Correlation between on-the-run discounts and other measures in levels. This table reports pairwise correlations between the two bill on-the-run discounts of Figure 1 and other Treasury-market measures, as in Table 2, using monthly averages in levels instead of first differences. $Spread_{26w}$ is the 26-week pre-reopening off/on spread; $Spread_{52w}$, $Spread_{2y}$, and $Spread_{10y}$ are the OIS-adjusted first off/on spreads for 52-week bills and 2- and 10-year notes; Note/bill is the note-bill spread; Noise is the Treasury noise measure; MOVE is the MOVE index. Numbers in brackets are p -values. All series are monthly averages of daily data in levels, January 2010 to December 2025. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	$Spread_{52w}$	$Spread_{2y}$	$Spread_{10y}$	Note/bill	Noise	MOVE
$Spread_{26w}$	0.62*** [0.00]	-0.13* [0.07]	0.28*** [0.00]	0.66*** [0.00]	-0.55*** [0.00]	-0.63*** [0.00]
$Spread_{52w}$		-0.34*** [0.00]	0.08 [0.25]	0.18** [0.01]	-0.41*** [0.00]	-0.50*** [0.00]
$Spread_{2y}$			0.07 [0.36]	0.01 [0.89]	0.24*** [0.00]	0.09 [0.21]
$Spread_{10y}$				0.38*** [0.00]	0.26*** [0.00]	-0.00 [0.98]

Table IA.2: The borrowing-demand measures before the reopening. Panels A and B report summary statistics of the two borrowing-demand measures for original 52-week Treasury bills on the days from each bill’s issue date to the day before the settlement of its 26-week reopening. Observations in these panels are bill-days. Panel A covers the Federal Reserve’s securities-lending operations. $BD_{i,t}^{Fed}$ is par submitted by dealers relative to par available for the security, in percent, first on operation days with bids and then on all operation days with days without bids counted as zero, as in the auction-cycle measures of Table 5; the indicator marks operation days with bids; par submitted and par available are in billions of dollars. Panel B covers Markit securities lending on days with lendable inventory. $BD_{i,t}^{Mkit}$ is the short-loan quantity relative to the actively lendable quantity, in percent, with days without loans counted as zero; the two quantities are in billions of dollars. Panel C reports the Federal Reserve borrowing-demand spread of the pairs, $BD_{i,t}^{Fed}$ for the seasoned security minus $BD_{i,t}^{Fed}$ for the newly issued security, averaged over the window, in percentage points, with days without bids counted as zero once a security is available for lending: the 52-week first off/on pair and the 26-week pre-reopening pair, one observation per auction cycle or reopening window, and the 2-year and 10-year first off/on note pairs, one observation per auction cycle. The bill series start in October 2019, when the Federal Reserve resumed holding bills; the note series start in 2010.

<i>Panel A. Federal Reserve securities-lending operations, October 2019 to 2025 (82 bills)</i>						
	Mean	SD	P25	Median	P75	N
$BD_{i,t}^{Fed}$ on days with bids (%)	26.10	42.44	0.88	8.06	31.85	6,896
$BD_{i,t}^{Fed}$, days without bids as zero (%)	19.12	38.28	0.00	1.55	19.44	9,330
Indicator for a day with bids	0.71	0.45	0.00	1.00	1.00	9,330
Par submitted on days with bids (\$ billion)	0.61	0.80	0.02	0.23	0.96	6,896
Par available on days with bids (\$ billion)	3.31	1.97	2.16	3.13	4.04	6,896
<i>Panel B. Markit securities lending, 2010–2025 (206 bills)</i>						
	Mean	SD	P25	Median	P75	N
$BD_{i,t}^{Mkit}$ (%)	38.07	36.74	12.96	30.96	57.39	26,285
Short-loan quantity (\$ billion)	0.38	0.42	0.08	0.22	0.52	26,285
Actively lendable quantity (\$ billion)	1.08	0.82	0.42	0.87	1.54	26,285
<i>Panel C. Federal Reserve borrowing-demand spread of the pairs (percentage points), window averages</i>						
	Mean	SD	P25	Median	P75	N
52-week first off/on pair, 2019–2025	5.31	14.49	-0.31	0.60	4.67	82
26-week pre-reopening pair, 2019–2025	53.44	46.23	16.59	39.53	94.63	82
2-year first off/on pair, 2010–2025	-4.30	27.71	-14.25	-0.99	6.03	122
10-year first off/on pair, 2010–2025	-33.83	25.73	-45.52	-28.48	-15.62	162

Table IA.3: Bid-ask spreads of the four bills in the two pairs. This table reports summary statistics of the bid-ask spreads of the bills that form the two on-the-run pairs of Table 1, the ask price minus the bid price relative to the price, in basis points of price. Panel A uses CRSP closing quotes and Panel B uses GovPX closing quotes; in GovPX, days with a negative quoted spread are dropped. The on-the-run and first off-the-run rows average daily observations within each four-week auction cycle of the 52-week bill; the pre-reopening and 26-week rows average within each four-week reopening window. The sample period is 2010–2025.

Panel A. Window-average CRSP bid-ask spreads (bps of price), 2010–2025

	Mean	SD	P25	Median	P75	<i>N</i>
On-the-run bid-ask spread	1.18	0.61	0.82	1.00	1.26	209
First off-the-run bid-ask spread	1.27	0.50	0.93	1.12	1.50	209
Pre-reopening off-the-run bid-ask spread	1.00	0.56	0.60	0.82	1.13	209
On-the-run 26-week bid-ask spread	0.48	0.14	0.38	0.44	0.53	209

Panel B. Window-average GovPX bid-ask spreads (bps of price), 2010–2025

	Mean	SD	P25	Median	P75	<i>N</i>
On-the-run bid-ask spread	0.89	0.49	0.58	0.71	1.03	208
First off-the-run bid-ask spread	1.33	0.96	0.68	1.00	1.65	208
Pre-reopening off-the-run bid-ask spread	1.40	1.13	0.67	1.06	1.78	208
On-the-run 26-week bid-ask spread	0.33	0.11	0.26	0.31	0.37	208

Table IA.4: Market liquidity, repo specialness, and on-the-run note spreads.

This table reports the statistics of Table 3 for the first off/on pairs of 2-year (Panel A) and 10-year (Panel B) Treasury notes: means, medians, and correlations for the OIS-adjusted on-the-run note spread, its repo-specialness spread, and its bid-ask differentials, averaged within auction cycles. Repo-specialness spreads are the on-the-run repo rate minus the off-the-run repo rate, so positive values indicate that the off-the-run note is more special. Bid-ask differentials are off-the-run minus on-the-run price bid-ask spreads from CRSP and GovPX. The correlation column reports correlations between cycle-to-cycle changes in each row variable and changes in the spread. The sample period is 2010–2025.

<i>Panel A. The 2-year first off/on spread, 2010–2025</i>					
	Mean	Median	N	Corr. with $\Delta Spread$	p -value
$Spread_{2y}$	0.29	0.22	192		
<i>Repo Specialness Spread</i> $_{2y}$	−2.14	−0.28	176	−0.20***	0.01
<i>Bid-ask Spread</i> $_{2y}$	0.07	0.00	192	0.08	0.30
<i>Bid-ask Spread</i> $_{2y}$ (GovPX)	0.17	−0.00	191	0.02	0.81
<i>Panel B. The 10-year first off/on spread, 2010–2025</i>					
	Mean	Median	N	Corr. with $\Delta Spread$	p -value
$Spread_{10y}$	1.13	0.85	193		
<i>Repo Specialness Spread</i> $_{10y}$	−27.42	−16.58	180	−0.32***	0.00
<i>Bid-ask Spread</i> $_{10y}$	−0.46	−0.07	193	0.23***	0.00
<i>Bid-ask Spread</i> $_{10y}$ (GovPX)	−0.72	−0.08	192	0.25***	0.00

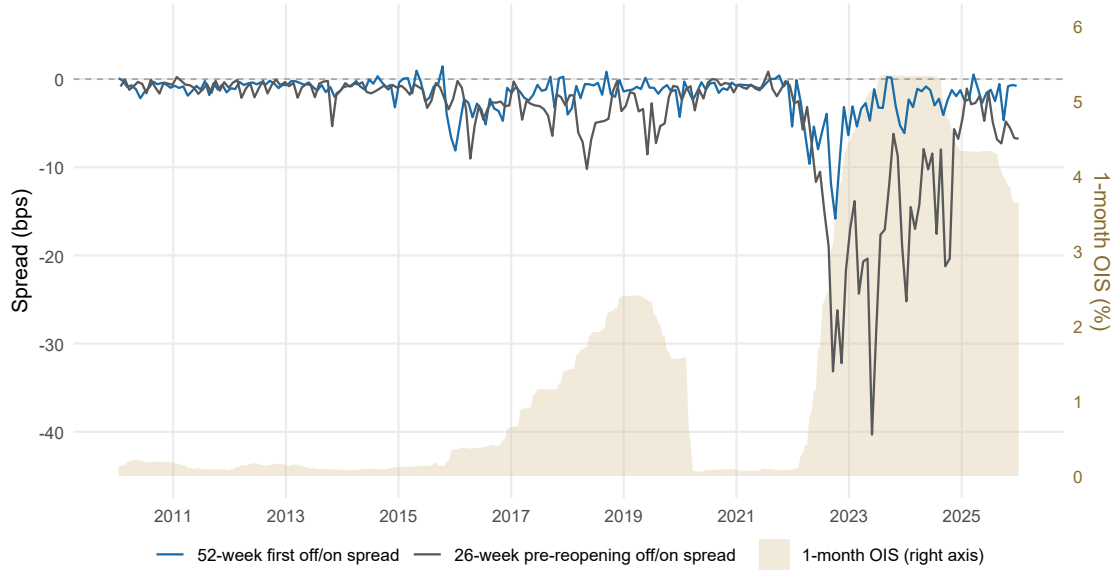


Figure IA.1: The on-the-run discounts and the level of short rates. This figure plots the two bill on-the-run discounts of Figure 1 together with the level of short-term interest rates. The blue line is the OIS-adjusted spread between the first off-the-run and the on-the-run 52-week bill, averaged within each four-week auction cycle; the grey line is the 26-week pre-reopening off/on spread, averaged within each four-week reopening window; both are in basis points on the left axis, and negative values mean that the newly supplied bill is cheap. The shaded area is the one-month OIS rate, in percent on the right axis, observed on the same dates as the spreads. Daily CRSP data from January 2010 to December 2025, 209 observations per series. Both discounts are widest when short rates are high: across the 2010–2015, 2016–2021, and 2022–2025 subperiods, the 52-week discount averages -0.9 , -1.6 , and -3.4 basis points and the 26-week pre-reopening discount -1.0 , -2.7 , and -12.8 basis points, while the one-month OIS rate averages 0.1 , 1.0 , and 4.1 percent. The correlation with the rate level is -0.28 for the 52-week and -0.71 for the 26-week pre-reopening discount; in cycle-to-cycle changes, both correlations are close to zero (-0.01 and -0.08).

IA.3 Additional Evidence on Mechanisms

This section provides additional evidence on market liquidity, investor demand, and the reopening mechanism. Table IA.5 reports trading volume for on-the-run and off-the-run Treasury securities. Figure IA.2 reports bid-ask spreads over the lifecycle of 52-week Treasury bills. Figure IA.3 reports money market fund holdings of 52-week Treasury bills by months to maturity. Figure IA.4 repeats the bill panel of Figure 5 for all Treasury bills. Table IA.6 repeats Table 4 with quadratic trends on each side of the reopening settlement. Table IA.7 repeats its borrowing-demand columns with the lendable amount held at its last pre-settlement value, so that the drop at delivery is not driven by a larger denominator.

Table IA.5: Trading volume by on-the-run status. This table reports average daily trading volume in billions of dollars for on-the-run and off-the-run Treasury securities from August 2017 through July 2018. Source: Brain et al. (2018).

Status	Coupons	Bills
On-the-Run	362	25
First Off-the-Run	31	8.3
Second Off-the-Run	7.7	7.6

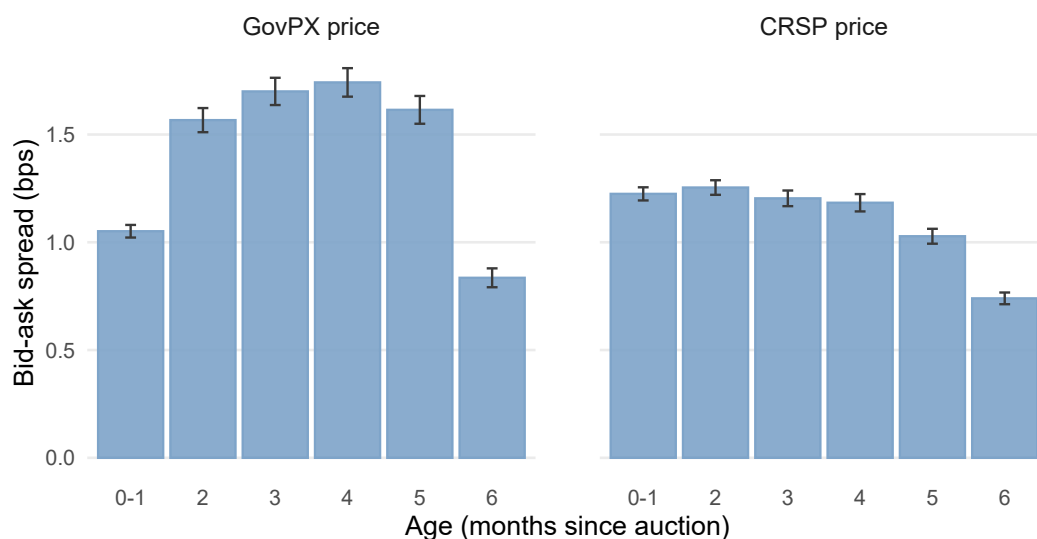


Figure IA.2: Bid-ask spreads over the 52-week bill lifecycle. This figure reports average bid-ask spreads for original 52-week Treasury bills by age in months since auction. The left panel uses GovPX quotes and the right panel uses CRSP quotes. Bid-ask spreads are measured in basis points of price and are computed as the ask price minus the bid price divided by the midpoint price. For GovPX bill quotes, discount-rate quotes are first converted into prices. Vertical bars denote 95% confidence intervals for the mean. The sample period is 2010–2025.

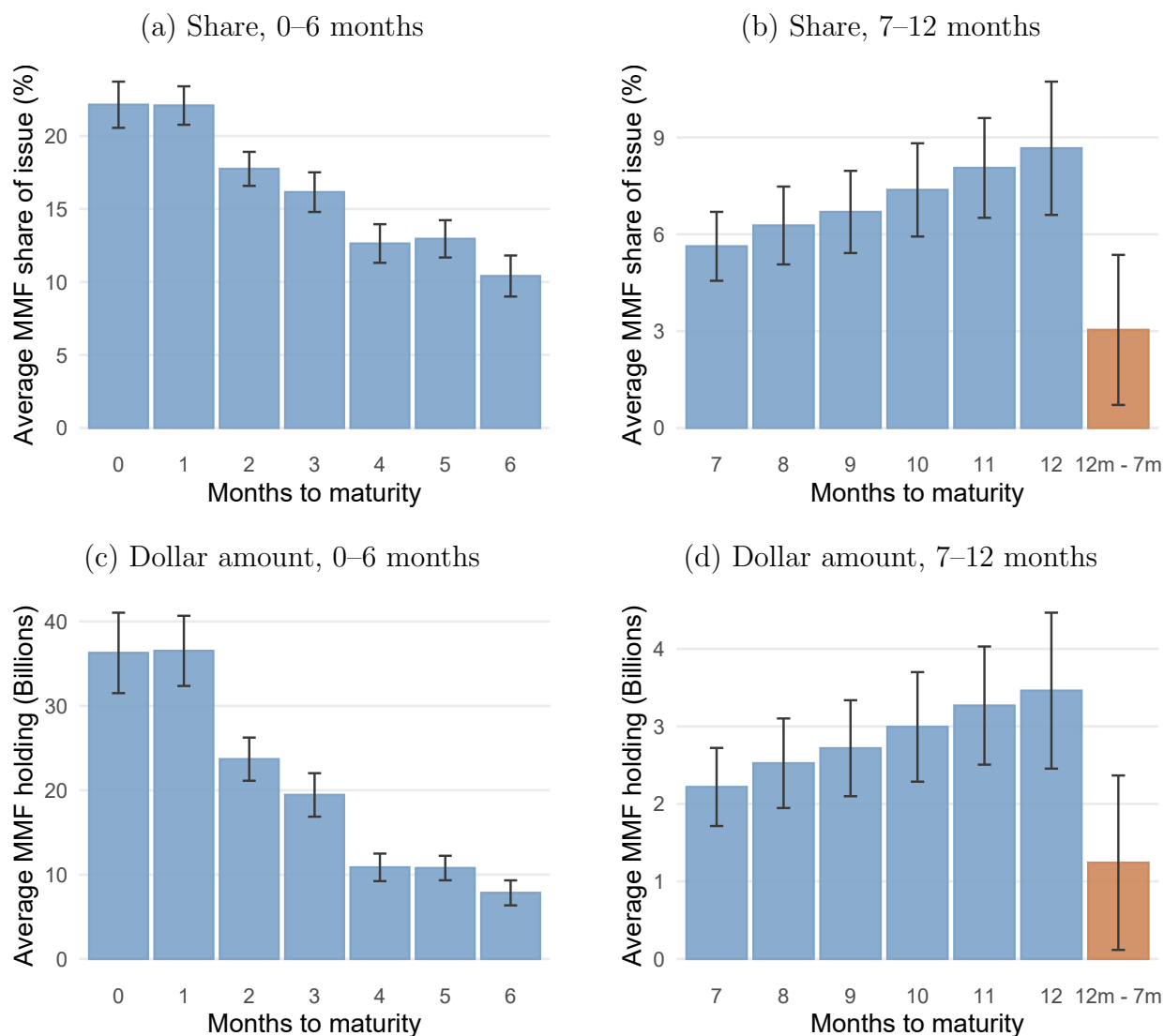


Figure IA.3: Money market fund holdings of 52-week Treasury bills. This figure reports money market fund holdings of 52-week Treasury bills by months to maturity, using month-end Crane holdings matched to Treasury issuance data. Panels (a) and (b) report the average fraction of each issue held by money market funds. Panels (c) and (d) report average dollar holdings in billions. Panels (a) and (c) focus on bills with 0 to 6 months remaining maturity, while Panels (b) and (d) focus on bills with 7 to 12 months remaining maturity. The final bar in Panels (b) and (d) reports the difference between the 12-month and 7-month bins. Vertical bars denote 95% confidence intervals for the mean.

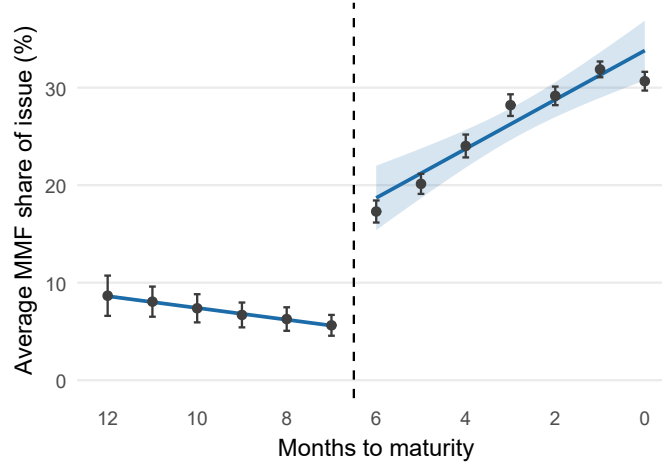


Figure IA.4: Money market fund holdings of all Treasury bills by remaining maturity. This figure repeats Panel (a) of Figure 5 using all Treasury bills instead of only securities issued as 52-week bills. Each dot is the average fraction of each issue held by money market funds, in percent of the amount outstanding, across issue-month observations in that months-to-maturity bin, using month-end Crane holdings matched to Treasury issuance data (1,132 securities, October 2011 to July 2025); whiskers denote 95% confidence intervals. The solid lines show separate weighted linear fits for the 7-to-12-month and 0-to-6-month segments, with shaded 95% confidence intervals, and the vertical dashed line marks 26 weeks to maturity. The bins above six months consist almost entirely of securities issued as 52-week bills, while the shorter bins also contain the shorter bill tenors, which money market funds hold intensively, so the levels in the 0-to-6-month segment are higher. The pattern is unchanged: the share falls as a security ages toward the 26-week mark and rises as maturity shortens thereafter.

Table IA.6: Discontinuity at the delivery of the reopening supply, quadratic specification. This table repeats Table 4 with a quadratic distance term on each side of the settlement: the regression adds $Distance_{i,t}^2$ and $Distance_{i,t}^2 \times \mathbb{1}(\text{Reopened})_{i,t}$ to Equation (2). The coefficient on $\mathbb{1}(\text{Reopened})_{i,t}$ remains the jump at the delivery of the reopening supply. Window, dependent variables, samples, fixed effects, and standard errors are as in Table 4.

	Specialness	$BD_{i,t}^{Fed}$	$BD_{i,t}^{Mkit}$
	(1)	(2)	(3)
$\mathbb{1}(\text{Reopened})_{i,t}$	-75.10*** (-8.47)	-69.08*** (-11.17)	-39.75*** (-16.30)
$Distance_{i,t} \times \mathbb{1}(\text{Reopened})_{i,t}$	-0.47 (-1.36)	-1.01*** (-2.66)	-0.09 (-0.44)
$Distance_{i,t}^2 \times \mathbb{1}(\text{Reopened})_{i,t}$	-0.04*** (-2.75)	-0.02 (-1.15)	0.002 (0.29)
$Distance_{i,t}$ (days)	1.22*** (7.73)	1.13*** (10.61)	0.27*** (6.66)
$Distance_{i,t}^2$	0.005*** (7.31)	0.005*** (9.54)	0.0002 (1.01)
Date FE	Yes	Yes	Yes
Sample	2012–2025	2019–2025	2010–2025
Observations	20,635	7,717	30,010
R ²	0.32	0.42	0.33
Adjusted R ²	0.19	0.28	0.22

Table IA.7: Discontinuity with the pre-reopening lendable amount as denominator. This table repeats Columns (2) and (3) of Table 4 with the denominator of each borrowing-demand measure held at its last observation before the settlement of the reopening. Before the settlement the measures are as in Table 4; from the settlement day on, par submitted in the Federal Reserve’s operations is divided by the par available on the last bid day before the settlement, and Markit’s short-loan quantity by the actively lendable quantity on the last day before the settlement. The lendable amounts themselves rise at the settlement, when the Federal Reserve receives its share of the reopening and lenders’ inventories grow, so this specification shows whether the drop in the ratios reflects a fall in borrowing or only the larger denominator. Window, fixed effects, standard errors, and samples are as in Table 4.

	$BD_{i,t}^{Fed}$	$BD_{i,t}^{Mkit}$
	(1)	(2)
$\mathbf{1}(\text{Reopened})_{i,t}$	-37.85^{***} (-6.85)	-17.12^{***} (-5.46)
$Distance_{i,t} \times \mathbf{1}(\text{Reopened})_{i,t}$	-0.11 (-0.36)	1.29^{***} (2.93)
$Distance_{i,t}$ (days)	0.32^{***} (8.93)	0.24^{***} (11.64)
Date FE	Yes	Yes
Sample	2019–2025	2010–2025
Observations	7,717	29,993
R ²	0.32	0.21
Adjusted R ²	0.14	0.09

IA.4 Other Robustness Tests

This section collects robustness tests for the main empirical results. Table IA.8 shows that the on-the-run discount is unchanged after excluding debt-limit episodes and bill pairs whose maturity dates straddle projected X-dates. Figure IA.5 plots the monthly GSW-adjusted 26-week first off/on spread from 1987 with the regime means from a Bai and Perron (1998) break test; the discount is negative and significant in every regime, smallest before July 2003, largest between 2003 and 2009, and without a further break after 2009. Table IA.9 reports the Markit and repo-specialness specifications corresponding to Table 5 on the cycles and windows with Federal Reserve lending data. Section IA.4.1 extends the borrowing-demand regressions to pairs of seasoned bills using the Federal Reserve and Markit measures in the two panels of Table IA.10.

Table IA.8: Robustness to debt-limit episodes and projected X-dates. This table reports auction-cycle summary statistics for the OIS-adjusted spread between the on-the-run 52-week bill and the first ($Spread_0^1$) or fourth ($Spread_0^4$) off-the-run bill. The baseline row uses the full 2010–2025 sample. The second row removes daily observations during the GAO acute-market-concern periods in 2011, 2013, 2015, 2017, 2018, 2021, and 2023. The third row removes an entire auction cycle if any trading day in that cycle falls in an acute period. The fourth row removes bill pairs whose maturity dates straddle a final or representative projected X-date during 2011–2025. The final row combines the auction-cycle and maturity-straddle exclusions. Mean spreads are expressed in basis points, with t -statistics based on variation across auction cycles reported in parentheses. Negative (%) is the percentage of auction-cycle spreads below zero, and N is the number of auction cycles.

Specification	$Spread_0^1$			$Spread_0^4$		
	Mean (t -stat.)	Negative (%)	N	Mean (t -stat.)	Negative (%)	N
Baseline	−1.77 (−11.8)	91.9	209	−5.52 (−13.9)	95.2	207
Exclude acute trading dates	−1.78 (−11.7)	93.1	203	−5.31 (−15.3)	95.0	201
Exclude overlapping auction cycles	−1.81 (−11.3)	93.2	190	−5.32 (−15.3)	95.7	188
Exclude X-date-straddling maturity pairs	−1.78 (−11.6)	91.9	197	−5.64 (−12.7)	95.7	163
Exclude overlapping cycles and straddling pairs	−1.82 (−11.0)	93.3	178	−5.41 (−14.6)	96.5	144

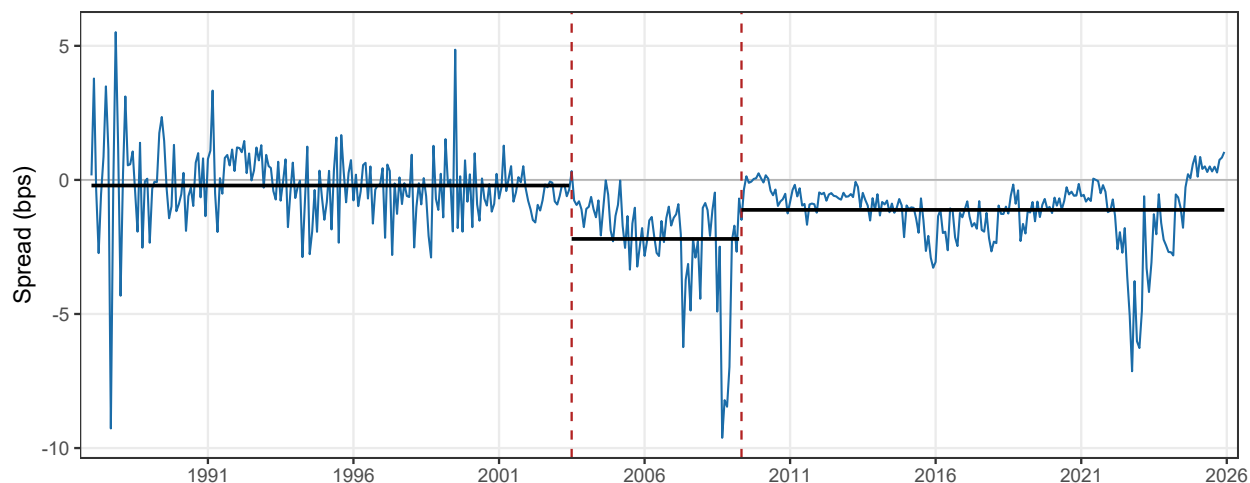


Figure IA.5: The 26-week on-the-run discount since 1987 and its regime shifts. This figure plots the monthly average of the GSW-adjusted spread between the first off-the-run and the on-the-run 26-week Treasury bill from January 1987 to December 2025, 468 months. Spreads are averaged within auction cycles and then within months; auction cycles with an absolute spread of 100 basis points or more are excluded. The vertical dashed lines mark the break dates selected by the Bai and Perron (1998) test for multiple shifts in the mean, with the number of breaks chosen by the BIC and a minimum segment length of 15 percent of the sample; the horizontal lines are the segment means. The breaks fall in July 2003 and May 2009, with 95 percent confidence intervals from August 2002 to January 2004 and from June 2008 to March 2012. The segment means are -0.21 , -2.20 , and -1.12 basis points, with Newey–West t -statistics of -2.1 , -5.5 , and -5.9 , so the discount is present in every regime and largest between 2003 and 2009. The level shifts at the two breaks are -1.99 basis points (t -statistic -4.9) and 1.08 basis points (t -statistic 2.5).

Table IA.9: Changes in the on-the-run discounts and relative borrowing pressure on the Federal Reserve sample. This table reports the Markit and repo-specialness specifications corresponding to Table 5 after restricting the sample to the auction cycles and reopening windows with Federal Reserve lending data. The Markit observation counts match the corresponding Federal Reserve columns of Table 5; the repo-specialness samples are slightly smaller where DTCC observations are unavailable. Panel A uses the change in the 52-week first off/on spread and Panel B the change in the 26-week pre-reopening off/on spread as the dependent variable. Columns (1) and (2) use the Markit short-loan share, expressed as a ratio, and Columns (3) and (4) the repo-specialness spread, in basis points. Controls and reported statistics are as in Table 5.

<i>Panel A. 52-week first off/on spread, $\Delta Spread_{52w,t}$, Federal Reserve sample</i>				
	(1)	(2)	(3)	(4)
$\Delta BD^{Mkit} Spread_t$	-2.22* (-1.97)	-2.93** (-2.31)		
$\Delta Specialness Spread_t$			-0.09*** (-6.74)	-0.09*** (-5.91)
$\Delta AS Spread_t$		1.61*** (3.08)		0.32 (0.59)
$\Delta AuctionDemand Spread_t$		-0.0000 (-0.0001)		-0.06 (-1.54)
$\Delta CDS Spread_t$		0.01 (0.04)		-0.10 (-0.57)
Market controls	No	Yes	No	Yes
Sample	2019–2025	2019–2025	2019–2025	2019–2025
Observations	81	80	75	75
Adjusted R ²	0.03	0.15	0.37	0.37
<i>Panel B. 26-week pre-reopening off/on spread, $\Delta Spread_{26w,t}$, Federal Reserve sample</i>				
	(1)	(2)	(3)	(4)
$\Delta BD^{Mkit} Spread_t$	-5.92** (-2.51)	-6.64** (-2.48)		
$\Delta Specialness Spread_t$			-0.03*** (-7.75)	-0.04*** (-9.10)
$\Delta AS Spread_t$		-0.05 (-0.03)		2.07 (1.37)
$\Delta AuctionDemand Spread_t$		-0.08 (-0.58)		0.10 (1.00)
$\Delta CDS Spread_t$		0.84 (0.88)		2.66*** (3.78)
Market controls	No	Yes	No	Yes
Sample	2019–2025	2019–2025	2019–2025	2019–2025
Observations	81	77	74	71
Adjusted R ²	0.06	0.01	0.45	0.53

IA.4.1 Relative Borrowing Demand among Seasoned Bills

This subsection moves beyond the on- and off-the-run comparison. If security-specific borrowing demand drives relative prices, the same relationship should appear between pairs of seasoned bills. For each off-the-run rank $i \geq 1$, $Spread_{i,t}^1$ is the OIS-adjusted yield of the $(i+1)$ -th off-the-run bill minus that of the i -th off-the-run bill. We define the Federal Reserve and Markit borrowing-demand spreads analogously as borrowing demand for the older bill minus borrowing demand for the younger bill, with each measure expressed as a ratio. A positive borrowing-demand spread should therefore be associated with a lower yield spread.

Panel A of Table IA.10 shows a negative and statistically significant relationship for all five adjacent pairs. A unit increase in the Federal Reserve borrowing-demand ratio for the older bill relative to the younger bill lowers the adjacent yield spread by 7.3 to 9.0 basis points. Equivalently, a one-percentage-point increase lowers the spread by approximately 0.07 to 0.09 basis points. Relative borrowing demand alone explains between 29% and 56% of the variation in auction-to-auction spread changes. Panel B provides similar evidence using the Markit short-loan share over the longer 2010–2025 sample. The coefficients are negative for all five pairs and statistically significant for the first four, although the explanatory power is lower, as for the on-the-run pairs in Table 5. Thus, the relationship between borrowing demand and relative prices extends across seasoned bills and is not specific to the on-the-run label.

Table IA.10: Borrowing demand and adjacent bill spreads. This table reports auction-cycle regressions for original 52-week Treasury bills. In Column i , the dependent variable is the auction-to-auction change in $Spread_{i,t}^1$, the OIS-adjusted yield of the $(i + 1)$ -th off-the-run bill minus that of the i -th off-the-run bill. The regressor is the corresponding change in the pair's borrowing-demand spread, borrowing demand for the older bill minus borrowing demand for the younger bill, constructed as in Table 5 and expressed as a ratio. In Panel A, borrowing demand is par submitted relative to par available in the Federal Reserve's securities-lending operations, and the sample consists of auction cycles with Federal Reserve lending data, 2019–2025. In Panel B, borrowing demand is Markit's short-loan quantity relative to actively lendable quantity on every day the security has lendable inventory, and the sample is all auction cycles, 2010–2025. The numbers in parentheses are t -statistics. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

<i>Panel A. Federal Reserve borrowing demand, 2019–2025</i>					
	$\Delta Spread_{1,t}^1$	$\Delta Spread_{2,t}^1$	$\Delta Spread_{3,t}^1$	$\Delta Spread_{4,t}^1$	$\Delta Spread_{5,t}^1$
	(1)	(2)	(3)	(4)	(5)
$\Delta BD Spread_{1,t}$	−7.76*** (−7.93)				
$\Delta BD Spread_{2,t}$		−8.42*** (−7.29)			
$\Delta BD Spread_{3,t}$			−8.64*** (−9.87)		
$\Delta BD Spread_{4,t}$				−8.95*** (−8.02)	
$\Delta BD Spread_{5,t}$					−7.32*** (−5.50)
Observations	81	81	78	77	74
Adjusted R ²	0.44	0.39	0.56	0.45	0.29
<i>Panel B. Markit short-loan share, 2010–2025</i>					
	$\Delta Spread_{1,t}^1$	$\Delta Spread_{2,t}^1$	$\Delta Spread_{3,t}^1$	$\Delta Spread_{4,t}^1$	$\Delta Spread_{5,t}^1$
	(1)	(2)	(3)	(4)	(5)
$\Delta BD Spread_{1,t}$	−1.06*** (−3.41)				
$\Delta BD Spread_{2,t}$		−1.01** (−2.55)			
$\Delta BD Spread_{3,t}$			−0.85** (−2.22)		
$\Delta BD Spread_{4,t}$				−1.22*** (−2.76)	
$\Delta BD Spread_{5,t}$					−0.66 (−1.58)
Observations	203	203	200	198	193
Adjusted R ²	0.05	0.03	0.02	0.03	0.01